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ROAD AND OFF-ROAD VEHICLE DESIGN

LAB REPORT

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General Performance of a Motor Vehicle

This exercise aims to study the maximum speed and acceleration of a vehicle. For this purpose, the point-mass model is applied: the car is represented by a point where all the mass of the vehicle is concentrated, and where all the forces that act on the system are applied.

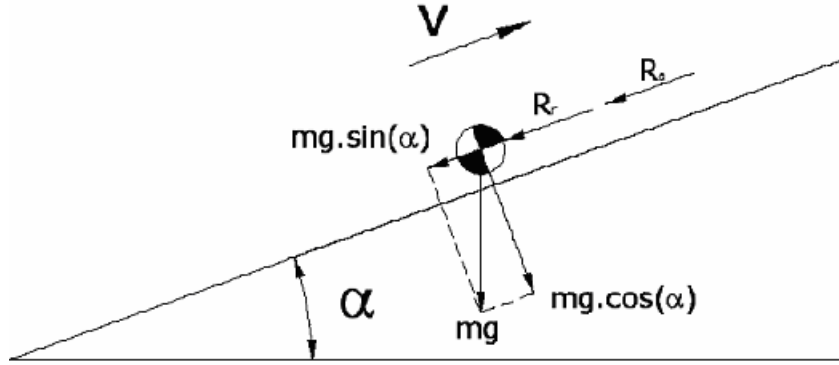


Figure 1: point-mass vehicle model

The vehicle is moving at a constant speed v on a straight path with ground inclination α . There are three forces acting on the car:

- Resistance due to upslope

$$R_i = P \cdot \sin(\alpha) = mg \cdot \sin(\alpha)$$

- Rolling resistance

$$R_0 = mg \cdot \cos(\alpha) \cdot (f_0 + f_2 \cdot v^2)$$

where the coefficients f_0 and f_2 consider the influence of the road surface; f_2 can be estimated, while f_0 refers to a smooth asphalt road:

$$f_2 = 0.5 \cdot 3.6^2 \cdot 10^{-6}$$

$$f_0 = 0.010$$

- Aerodynamic resistance

$$R_a = \frac{1}{2} \rho C_x S v^2$$

where $\rho = 1.2 \text{ kg/m}^3$ is air density, S is the frontal area of the car and C_x is the drag coefficient.

- The forward pushing force from the car F_x .

The resultant R_R of resistance forces then becomes:

$$R_R = R_i + R_0 + R_a = mg \cdot \sin(\alpha) + mg \cdot \cos(\alpha) \cdot (f_0 + f_2 \cdot v^2) + \frac{1}{2} \rho C_x S v^2$$

The power requirement can be calculated by multiplying the resultant force by the vehicle's velocity:

$$N_R = R_R \cdot v$$

Knowing the engine torque curve $C_m(\omega)$ and available power $N_d(\omega)$, the available power at wheels $N_e(\omega)$, can be determined:

$$N_e = N_d \eta_{ci} \eta_p$$

Where η_{ci} is the gearbox efficiency of i -th gear, η_p is the final drive efficiency.

The relation between vehicle speed v and engine rotational speed n [rpm] is as follows:

$$v = n \cdot \frac{2\pi}{60} \tau_{ci} \tau_p r$$

where τ_{Ci} is the i -th gear ratio, τ_p is the final drive ratio and $r = r_0 - \Delta r$ is the deformed radius of the pneumatic tyre. The tyre radial deformation can be computed from the tyre's radial stiffness: $\Delta r = \frac{mg}{4k_t}$

The vehicle object of this study is the 2020 BMW 330i, equipped with a 2.0-litre turbocharged 4-cylinder petrol engine and an 8-speed automatic transmission.



Figure 2: 2020 BMW 330i

Name	Symbol	Value	Units
Curb weight	m_{curb}	1505	kg
Wheel moment of inertia	J_w	0,6	kgm ²
Engine moment of inertia	J_m	0,19	kgm ²
Transmission moment of inertia	J_t	0,07	kgm ²
Frontal area	S	2,22	m ²
Drag coefficient	C_x	0,24	
Engine displacement	C	1,998	l
Differential efficiency	η_P	0,95	
Gearbox efficiency	η_{Ci}	0,98	
1st gear ratio	τ_I	0,190476	
2nd gear ratio	τ_{II}	0,297619	
3rd gear ratio	τ_{III}	0,460829	
4th gear ratio	τ_{IV}	0,581395	
5th gear ratio	τ_V	0,757576	
6th gear ratio	τ_{VI}	1	
7th gear ratio	τ_{VII}	1,219512	
8th gear ratio	τ_{VIII}	1,5625	
Final drive ratio	τ_p	0,355872	
Undeformed tyre radius	r_0	3284	mm
Tyre radial stiffness	k_t	200000	N/m

Table 1: 2020 BMW 330i data

As shown in the torque curve, the engine is capable of delivering 400 Nm from 1500 to 4500 rpm; after that, the power is constant at 190 kW up to the rev-limiter at 6500 rpm.

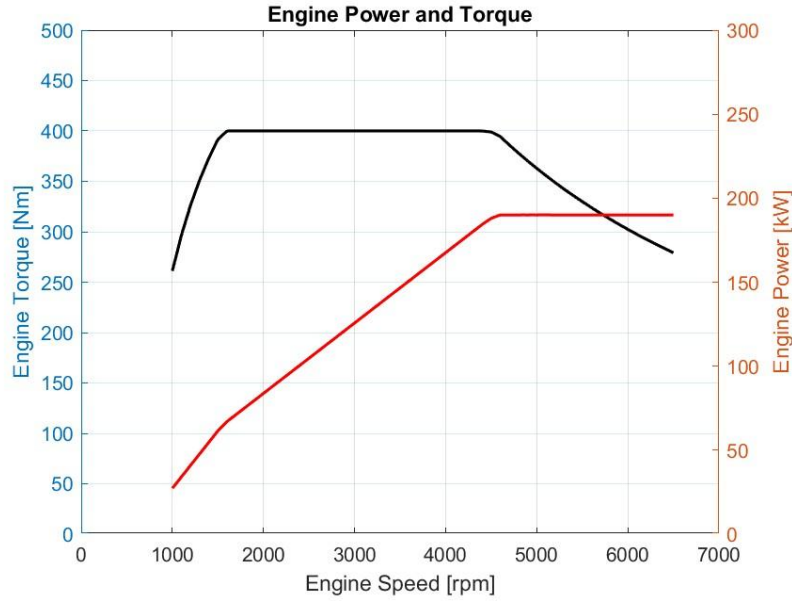


Figure 3: Engine power and torque curve

Initially, the vehicle will be studied in running order: only the driver will be accounted for, and the vehicle is supposed to have half a tank of fuel and 5 kg of luggage. The fuel density is $\rho_{fuel} = 770 \text{ kg/m}^3$ and the fuel tank volume is $V_{tank} = 59 \text{ l}$.

$$m_{running} = m_{curb} + m_{driver} + m_{luggage} + m_{fuel} = 1505 + 70 + 5 + 22.7 = 1602.7 \text{ kg}$$

$$m_{fuel} = 0.5 \cdot \rho_{fuel} \cdot V_{tank} = 22.7 \text{ kg}$$

The following part of this exercise will focus on studying the proper gear ratios for the optimal vehicle general performance. The resistance power and engine power can be plotted in order to compute the maximum vehicle speed on a flat surface and to tweak the ratio of the highest gear. After accounting for the drivetrain efficiency losses, the power available at the wheels is about 176,9 kW, so the maximum speed available is 263,5 km/h (this value is compatible with the real car, which is electronically limited at 250 km/h).

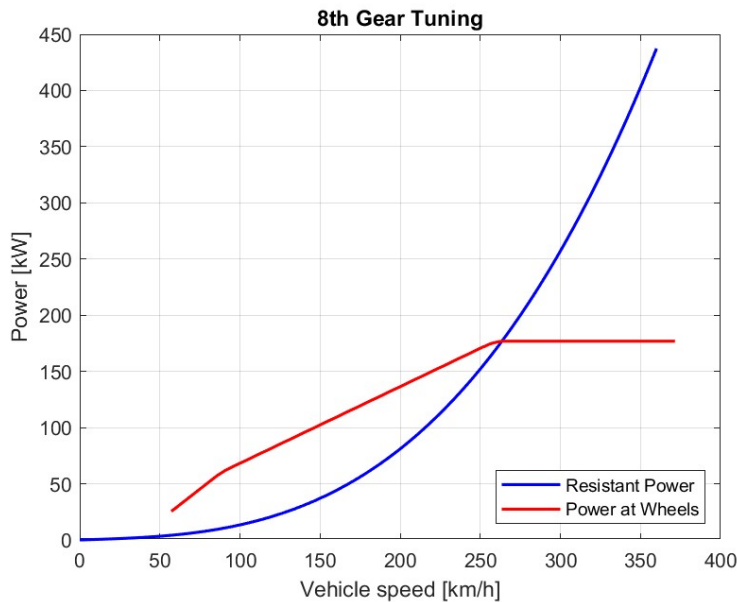


Figure 4: 8th gear power delivery

By tuning the 8th gear ratio, the power curve can be translated horizontally in the graph to optimize the engine revs at which the maximum vehicle speed is met. Due to this engine peculiar power delivery, the 8th gear ratio has been imposed at $\tau_{VIII} = 1,38$ in order to meet this condition at the earliest in the rev range: the transmission has many gears so the power availability across a wide range of vehicle's speed is ensured. With this configuration the gear ratios are as spread out as possible in order to provide best elasticity to the power train.

To compute the ratio of the lowest gear, the car must climb a slope of 45%.

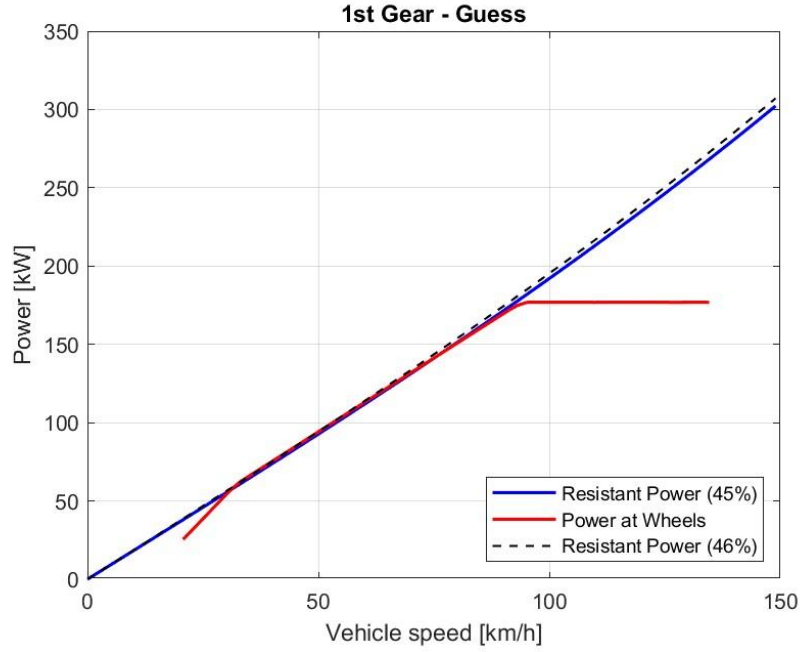


Figure 5: 1st gear transmission ratio guess

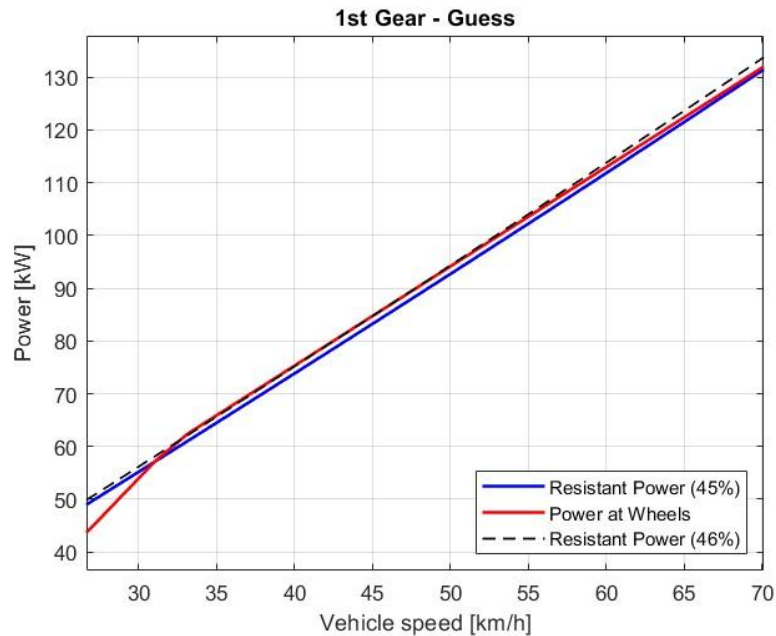


Figure 6: Particular of 1st gear transmission ratio guess

With a 1st gear ratio of $\tau_{I,guess} = 0,50$ the power available at the wheels perfectly matches the power needed to climb a 46% slope between 33 km/h and 55 km/h, so it is perfectly capable of climbing a slope of 45%. When running in first gear, the vehicle must be suitable for traffic jams and low speed manoeuvring, so when the engine runs at minimum rpm (1000) the vehicle has to run between 7 and 12 km/h. With the previous 1st

gear ratio, the speed in this condition is $v_{min} = 20,71 \text{ km/h}$, so far off the suggested speed range, it has to be decreased. To find a better suited ratio for low speed, a target minimum velocity of $v_{min} = 9 \text{ km/h}$ has been imposed and a proper 1st gear ratio has been computed: $\tau_I = 0,2173$.

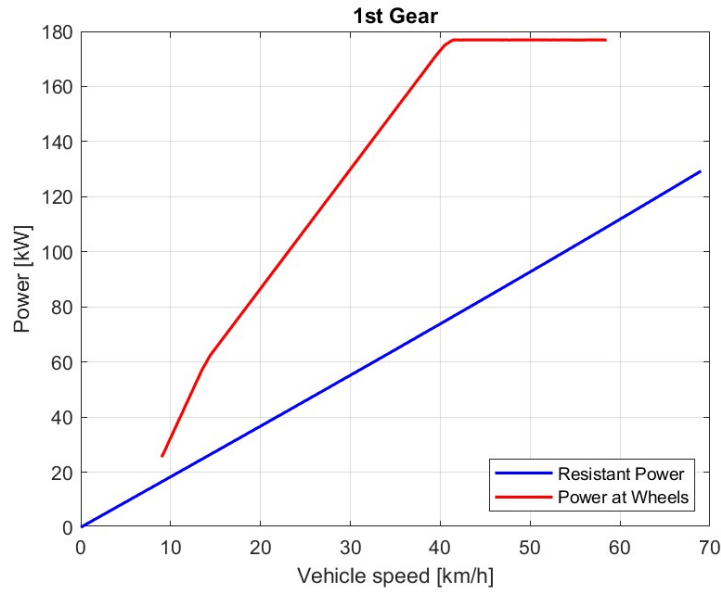


Figure 7: 1st gear power delivery

With this transmission ratio the car can easily overcome slopes of 45% (and higher). In addition to this, a lower gear ratio allows to achieve maximum power at lower speed and to improve acceleration.

Now that the minimum and maximum gear ratios are set, all the intermediate gears ratios can be computed according to the formulas reported below (geometric progression): in the case of the BMW, the number of gears is $n = 8$.

$$\tau_t = \frac{\tau_{min}}{\tau_{max}} = \frac{\tau_I}{\tau_{VIII}} = const.$$

$$\tau_t^{\frac{1}{n-1}} = \frac{\tau_{i-1}}{\tau_i}$$

Gear	Real ratio	Computed ratio
1	0,1905	0,2173
2	0,2976	0,2830
3	0,4608	0,3685
4	0,5814	0,4799
5	0,7576	0,6249
6	1,0000	0,8138
7	1,2195	1,0597
8	1,5625	1,3800

Table 2: Gear ratios

The real first gear ratio is comparable to the computed one, being about 10% smaller. The real 8th gear ratio is bigger than the computed one, so the real vehicle probably is not capable of reaching the maximum speed in 8th gear. This choice is probably due to high-speed efficiency: with a taller 7th and 8th gears the real car can travel at highway speeds with lower engine revs, thus lowering fuel consumption. Overall, the real car has more spread-out gear ratios, fully exploiting the relatively high number of gears.

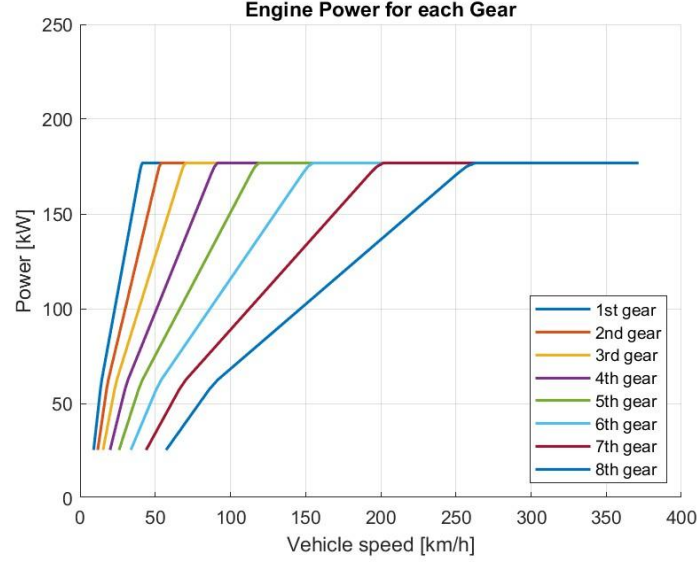


Figure 8: Power delivery of each gear

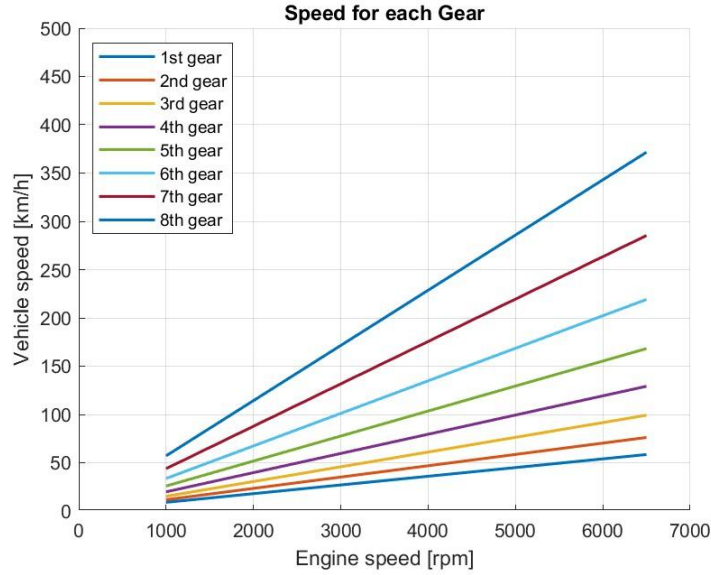


Figure 9: Speed profile of each gear

Finally, the acceleration of the vehicle is studied. The vehicle's performance will be evaluated by the time required to accelerate from 70 to 120 km/h in the highest gear. To calculate this, the useful power can be computed as follows:

$$N_u = N_e - N_r$$

And so from N_u the traction force R_u can easily be calculated:

$$R_u = R_e - R_r = m \cdot a$$

$$a = \frac{R_u}{m}$$

In this case the inertia of wheels, transmission and engine cannot be neglected, so m is an “equivalent mass” term, accounting for the inertia of those components.

$$m = m_{running} + 4 \cdot \frac{J_{wheel}}{r^2} + \frac{J_{trans}}{r^2 \tau_p^2} + \frac{J_{engine}}{r^2 \tau_p^2 \tau_{VII}^2}$$

The acceleration time can be calculated by discretizing the speed range of interest. The time required to increase the velocity Δv assuming constant acceleration $\bar{a}$ is calculated as $\Delta t = \Delta v / \bar{a}$. The total time can be calculated by summing the time Δt over the complete speed range.

$$\Delta t_{70-120} = \sum_i \frac{\Delta v_i}{\bar{a}}$$

The following result is obtained: $\Delta t_{70-120} = 12,1176$ s. This is a plausible value considering that the real car accelerates from 80 km/h to 120 km/h in 10,0 s in 7th gear.

The same study is now carried out for a full load case: the vehicle is supposed to carry all the allowable passengers (5 in this case), 15 kg of luggage per person and a full tank of fuel.

$$m_{full} = m_{curb} + 5 \cdot m_{passenger} + 5 \cdot m_{luggage} + m_{fuel} = 1505 + 5 \cdot 70 + 5 \cdot 15 + 45.4 = 1975.4 \text{ kg}$$

$$m_{fuel} = \rho_{fuel} \cdot V_{tank} = 45.4 \text{ kg}$$

When travelling on a flat surface at constant velocity, the weight increase has no effect on the maximum speed of the vehicle, so the maximum speed and 8th gear ratio values are identical to the previous case. The weight increase mostly affects the power needed to climb a slope, so in this case the guess for the 1st gear ratio is slightly decreased: $\tau_{1,guess} = 0,41$.

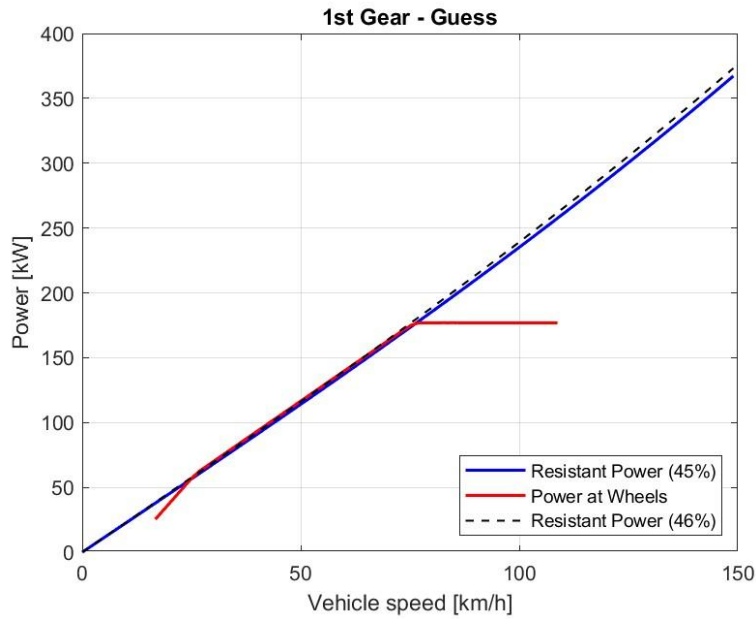


Figure 10: 1st gear transmission ratio guess - full load

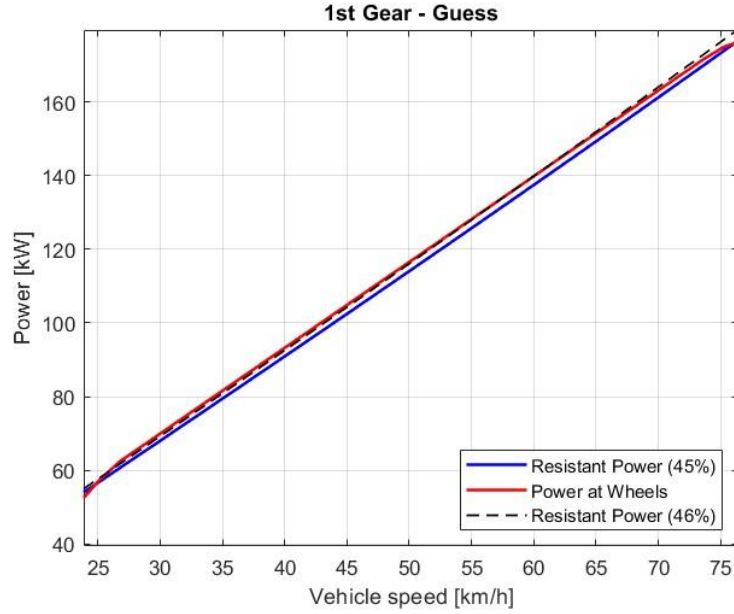


Figure 11: Particular of 1st gear transmission ratio guess - full load

The same procedure has been followed as per the case of the vehicle in running order. Once again, this gear ratio is not suitable for low-speed manoeuvres, so the value of $\tau_I = 0,2173$ will be maintained. Since both the 1st and 8th gear ratio remain unchanged, all the others will not undergo any adjustment either. At the end, the difference in weight did not cause any difference in the gear ratio final computation.

This can not be said for the acceleration performance of the vehicle: the increased inertia of the car results in more time needed to accelerate from 70 km/h to 120 km/h. the result of the calculation is, in fact, $\Delta t_{70-120} = 14,5989$ s.

Ride Comfort

Question 1: most comfortable car and relevant parameters

The objective was to determine which car among Alfa Romeo Giulietta, Alfetta and 159 is the most comfortable. The first one, Alfa Romeo Giulietta, is from 1960, while the other two, Alfetta and 159, are respectively from 1984 and 2007. So, investigating the differences among them also means to have a look on how the car comfort has changed along the last 60 years.

In the next image is visible the quarter car model considered to evaluate the comfort and in the table are reported the corresponding values of damping, masses and stiffnesses for the three cars.

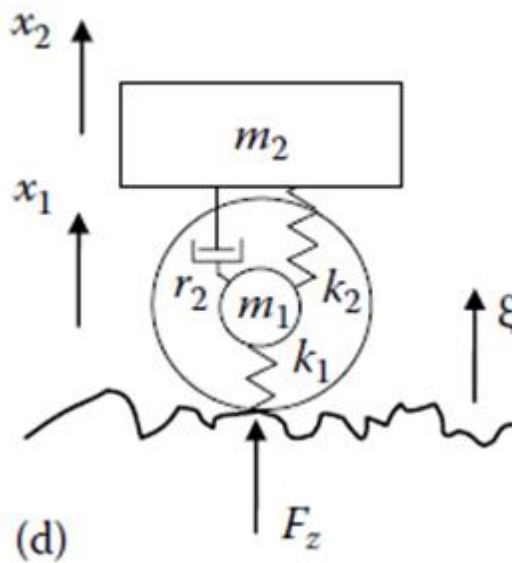


Figure 12: Quarter vehicle model

	Giulietta	Alfetta	159
m_1 [kg]	25	30	33
m_2 [kg]	200	270	350
k_1 [N/m]	100000	150000	270000
k_2 [N/m]	10000	20000	28000
r_2 [Ns/m]	1500	2500	5200

Table 3: Cars data for quarter vehicle model

By means of the following formulas the discomfort index has been computed for each one of the cars and then; by comparing the discomfort indexes of the three cars, it has been possible to assess which was the most comfortable one.

$$\sigma_{\ddot{x}}^2 = \frac{1}{2} A_b v \bar{\sigma}_{\ddot{x}2}^2$$

$$\bar{\sigma}_{\ddot{x}2}^2 = \frac{1}{m_2^2} \left(\frac{(m_2 + m_1)k_2^2}{r_2} + k_1 r_2 \right)$$

In the first formula A_b depends on road irregularities and has been considered equal to 1.4×10^{-5} m for the case of a normal road and to 1.0×10^{-4} for the case of a rough road; v , instead, is the vehicle speed, in this case 30 m/s.

	Giulietta	Alfetta	159
σ_{x2} (normal road)	0.930726	1.103865	1.582987
σ_{x2} (rough road)	2.487469	2.950204	4.230711

Table 4: σ_{x2} for normal and rough roads

In the table are visible the results, and so the discomfort indexes of the three cars both for the normal and for the rough road. The most comfortable one is the Giulietta, followed by the Alfetta and then, as last, by the 159. To understand why we observe such behaviour we need to run a sensitivity analysis in order to assess which of the parameters that influence the discomfort index are the most relevant. In the diagram are reported the percentual variation of the discomfort index of the Alfetta as function of a 50 % variation of the quarter car model parameters (m_1 , m_2 , k_1 , k_2 , r_2).

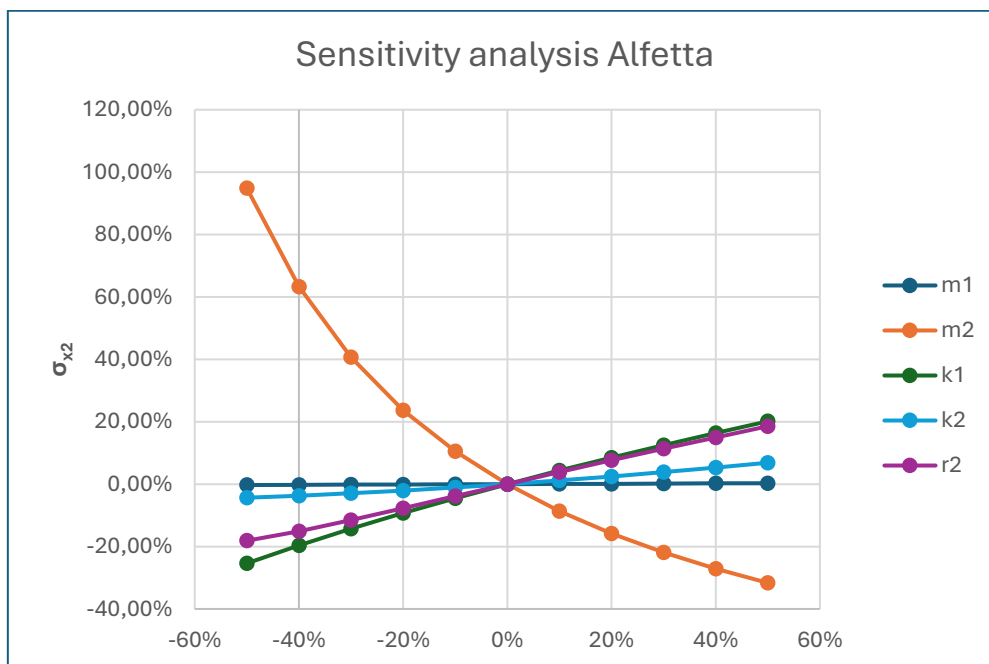


Figure 13: Sensitivity analysis for Alfa Romeo Alfetta

As it is clear the most influential of the parameters is the sprung mass m_2 : an increase in sprung mass causes a reduction of σ_{x2} ; but despite its increment passing from older to most recent models, we see an increase of the discomfort index. This can be explained by considering the increase of k_1 (tire radial stiffness) and r_2 (damping) that, even if according to sensitivity analysis should be less influential, show a much higher percentage variation among the three cars respect to the one of the masses, as visible in the table, causing consequently an increase of σ_{x2} . The same can be said for k_2 , even though considering the sensitivity analysis it is probably less important. Eventually also the effect of the unsprung mass m_1 should be considered, but it is of much lower intensity respect to the other and likely negligible.

	Giulietta	Alfetta	159
m_1	0%	20%	32%
m_2	0%	35%	75%
k_1	0%	50%	170%
k_2	0%	100%	180%
r_2	0%	67%	247%

Table 5: Parameters' effect on σ_{x2}

Question 2: Reason of the differences

The increase in discomfort is evident from Table 4, but the reasons behind that are not so trivial. The first cause that could explain why we observe such an evolution of the discomfort index over the years might be related to the road conditions; in fact, nowadays, most of the roads are paved and it is much more unlikely to ride on gravel or dirt roads than it was in the 1960s. The values of σ_{x2} show that for Alfetta and 159 running on white road would be close to impossible. In general, it can be said that the better quality of the roads doesn't require anymore such a low discomfort index.

Another explanation can be found in the characteristics of modern vehicles: nowadays cars are faster, more dynamically capable and heavier. A stiffer suspension system is necessary to ensure higher stability, controllability and safety and to sustain the increasing weights. This change is also very evident when comparing cars of similar class from different time period: the evident roll of older cars in corners is unseen in modern vehicle and is a consequence of all the design choices discussed above.

Design of a MacPherson Suspension

Introduction

The design of a vehicle suspension system is a complex engineering task that involves a thorough understanding of vehicle dynamics, structural integrity, and manufacturing processes. This exercise delves into the development of a front McPherson suspension system for a front-wheel-drive vehicle, highlighting the critical steps and considerations involved in the design process.

The MacPherson suspension, known for its simplicity and effectiveness in front-wheel-drive configurations, requires precise engineering to ensure optimal performance and safety. The primary objective of this exercise is to create a comprehensive assembly drawing of the suspension and a detailed production drawing of its lower arm. This involves defining the positions of the kinematic points of the suspension according to specific requirements related to the vehicle's operating conditions and spatial constraints. To begin with, the positions of the kinematic points must be determined to meet the given requirements on the suspension's operating conditions while complying with the available space.

These requirements are specified in terms of characteristic angles of the suspension, such as camber and steering angles, and the constraints are defined by the allowable locations for the joints. This necessitates a careful balance between achieving the desired kinematic objectives and fitting within the physical limits of the vehicle's layout. A critical aspect of the suspension design is ensuring the structural safety of the system. This involves estimating the forces acting on the suspension and computing the load on each component. Special attention is given to the lower arm, which must be sized accurately to withstand the computed forces. This includes considering the most stressed zones of the component and ensuring that the design can endure the dynamic loads experienced during operation.

Moreover, the manufacturing process plays a significant role in the design of the lower arm. Selecting the appropriate manufacturing techniques is essential to achieve the required strength and durability. The exercise encourages students to apply various manufacturing and structural optimization techniques learned during their studies to refine the design.

Kinematic optimization

The design process assumes that a prior dynamic study of the vehicle has been conducted, establishing specific requirements for the angles based on wheel travel; those targets have been reported in the following table:

	Nominal position	Full compression (80 mm)	Full extension (95 mm)
Camber angle	-1°	-0°26'30"	+0°28'24"
Steering angle	-0°03'17"	-0°46'22"	+0°26'10"

Table 6: Suspension's angles targets

Additionally, the vehicle's layout imposes strict constraints on the available space for the suspension system. These spatial constraints are translated into allowable coordinate ranges for the suspension joints, ensuring that the design adheres to the defined limits. Those coordinates are expressed in the vehicle's reference system that has origin in a point belonging to the mean plane of the vehicle, x axis along the longitudinal direction of the vehicle rearwards, z axis vertical upward and y axis as to create a right-handed system. In the following figure, the reference system and the reference points of the joints are shown.

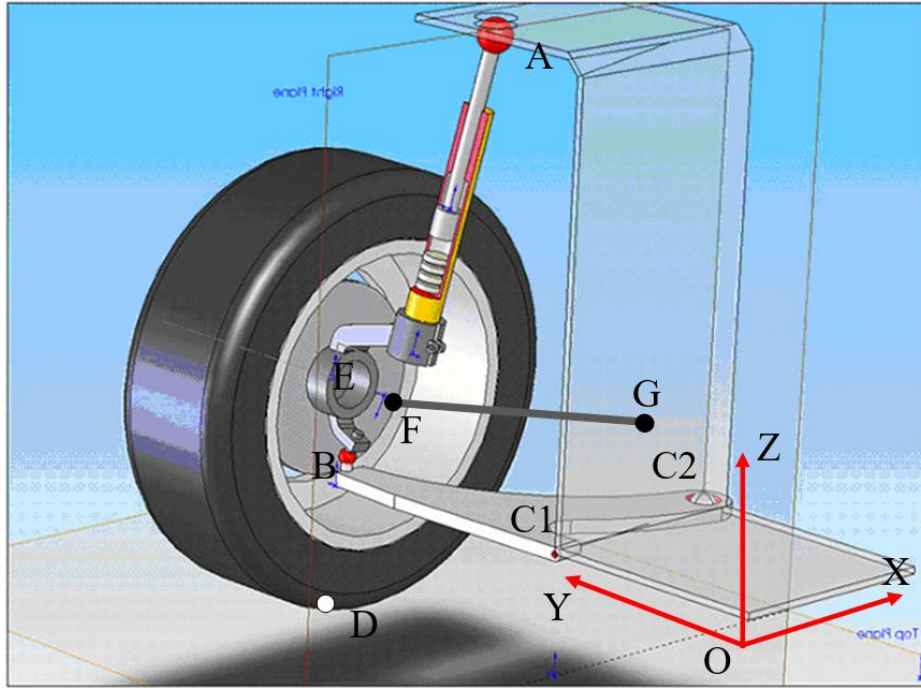


Figure 14: MacPherson suspension example

The inclination of the damper is $3^{\circ}32'30''$ forward and $2^{\circ}30'48''$ outward in the nominal configuration of the suspension. The reference tyre size is 205/60R15. Points description, fixed coordinates and allowable ranges for the design variables are reported in the following table (point C2 is defined relatively to point C1).

Point	Description	Coordinates (x,y,z) [mm]
A	Rotation centre of the upper bushing	(31.8 593 604)
B	Joint between wheel hub carrier and lower arm	(-9.10, 715 $\div$ 735, -50 $\div$ -70)
C1	Joint between chassis and lower arm	(-10, 390 $\div$ 410, -45 $\div$ -65)
C2	Joint between chassis and lower arm	(C1 _x +321, C1 _y , C1 _z +4,6)
D	Tyre-road contact point	(-1.6, 773.5, -226.8)
E	Wheel centre	(-1.6, 768.4, 65.1)
F	Steering joint on the wheel hub carrier	(136.9, 682.7, 88.5)
G	Steering joint on the chassis (centred steer)	(170, 300 $\div$ 320, 86)
O	Reference system origin	(0,0,0)

Table 7: Points' descriptions and coordinates

The design variables for this optimization problem are:

- Point B: coordinates y e z
- Point C1: coordinates y e z
- Point G: coordinate y

The optimization method adopted for this problem is an Artificial Neural Network that has been trained with a set of kinematic simulation of the suspension: it takes as an input the variable coordinates of the points and it outputs the steer and camber as a function of the wheel travel. The output of the neural network has been used to set up an optimization algorithm on MATLAB with the objective of minimizing the mean square error between the target values and the steer and camber functions predicted by the ANN. The algorithm used to minimize the error is an iterative method (SQP) suited for non-linear, constrained optimization problems. This

procedure gives very precise values that have little relevance in an engineering environment, so the final value of the points coordinates has been rounded.

	Constraints/target	Optimized values	Final values	Units
C1(y)	390 ÷ 410	391,9575540371	392,0	mm
C1(z)	-45 ÷ -65	-45,0000000000	-45,0	mm
G(y)	300 ÷ 320	314,5118661417	314,5	mm
B(y)	715 ÷ 735	715,0000000000	715,0	mm
B(z)	-50 ÷ -70	-51,0727152445	-51,0	mm
Camber (compressed)	-0°26'30''	-0°26'30''	-0°27'23''	Degrees
Camber (nominal)	-1°	-1°	-1°0'12''	Degrees
Camber (extended)	+0°28'24''	+0°28'24''	+0°27'40''	Degrees
Steering (compressed)	-0°46'22''	-0°46'22''	-0°45'4''	Degrees
Steering (nominal)	-0°03'17''	-0°03'17''	-0°03'30''	Degrees
Steering (extended)	+0°26'10''	+0°26'10''	+0°25'57''	Degrees

Table 8: Points optimization results

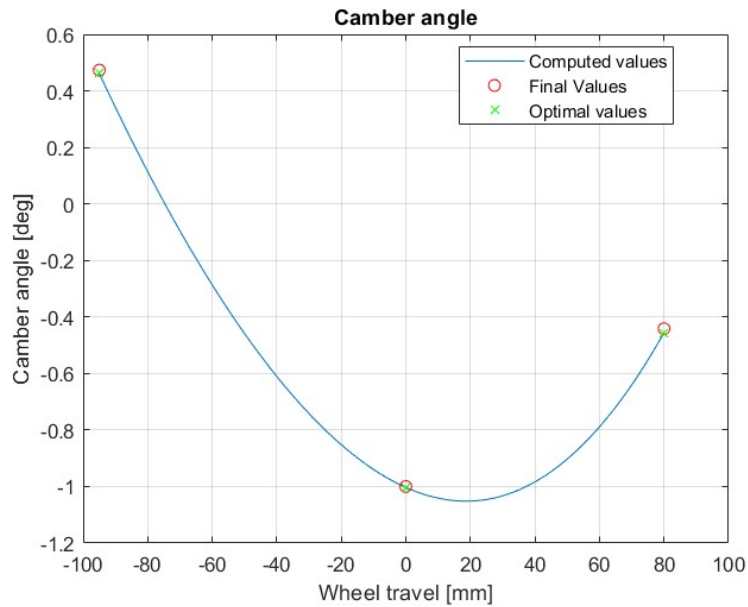


Figure 15: Camber angle as function of wheel travel

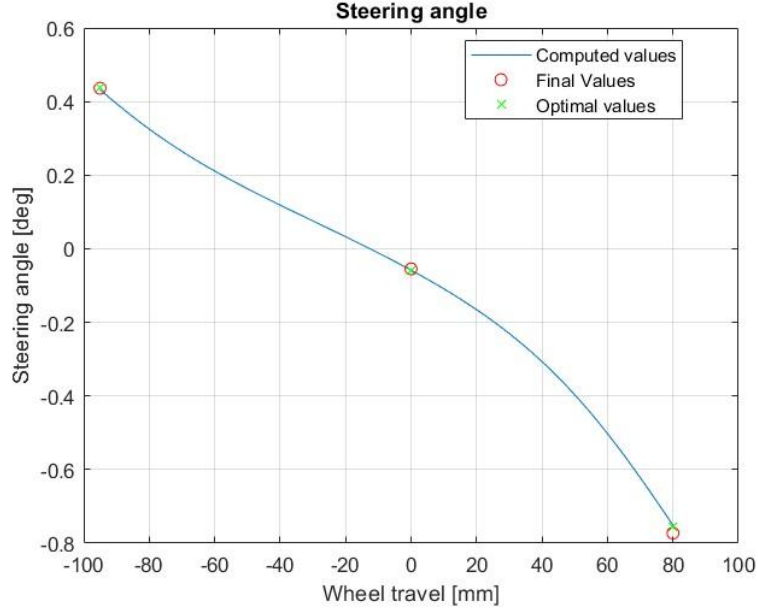


Figure 16: Steering angle as function of wheel travel

Force computation

In order to design the lower arm, the definition of the forces acting on the suspension system is required. The simultaneous application to the tyre of forces along the three directions allows the definition of the most demanding situation for the structural safety of the suspension. In the vertical direction a force Z equal to k times the vertical load acting on the tyre is considered. The k value considers the dynamic effects due to road unevenness or bumps.

$$Z = k \cdot F_{z,stat}$$

For the static safety of the suspension, a k coefficient equal to 2.5 has to be considered. The lateral force can be computed as

$$Y = \mu_Y \cdot Z \quad (\mu_Y = 0.7)$$

This condition refers to the hitting of a bump when the vehicle is negotiating a turn at high lateral acceleration. Finally, a longitudinal force (traction or braking) is added as ($\mu_X = 1.2$):

$$X = \mu_X \cdot Z$$

Starting from the defined condition, by imposing the equilibrium for the rotation around the hinge that corresponds to the dome, the forces acting on the lower arm can be computed. In particular, the following simplified scheme has been considered, with the lower arm approximated as a rod in the vertical plane.

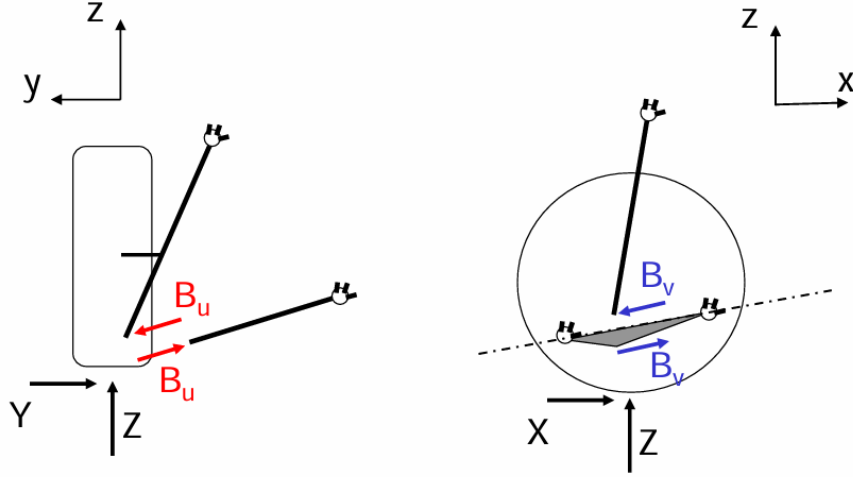


Figure 17: Forces exchanged between lower arm and hub carrier

Now, considering β_1 the inclination angle of the lower arm in the yz plane and β_2 the one in the xz plane, the following equations can be derived from the equilibrium equations.

$$B_u = \frac{Y * |z_D - z_A| - Z * |y_D - y_A|}{\cos(\beta_1) * |z_B - z_A| - \sin(\beta_1) * |y_B - y_A|}$$

$$B_v = \frac{X * |z_D - z_A| - Z * |x_D - x_A|}{\cos(\beta_2) * |z_B - z_A| - \sin(\beta_2) * |x_B - x_A|}$$

The values of B_u and B_v are respectively $5,887 * 10^3$ N and $1,409 * 10^4$ N. Once known them, the reactions at the bushings can be derived by simple equilibrium equation for the lower arm. To do that we must consider the axial stiffness of the bushings, that being the same for the two of them in our case, leads to have an equal distribution of the axial force among them.

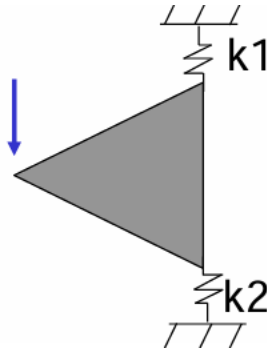


Figure 18: Stiffness of lower arm's bushings

So, the values for the axial force (T) and the normal one (N) at the bushings are:

$$T_{C1} = T_{C2} = 7,042 * 10^3 \text{ N}$$

$$N_{C1} = -8.306 * 10^3 \text{ N}$$

$$N_{C2} = 1.419 * 10^4 \text{ N}$$

Once also the reactions on the bushings have been computed the internal action on the lower arm can be computed and required characteristic for the bushings can be derived.

Lower arm design process

The selected manufacturing process for the production of the lower arm is die casting. This suspension system is supposed to belong to a high-production number vehicle (MacPherson layout is very common in front wheel drive hatchbacks) and this process allows to achieve high volumes, fast production time while maintaining a good value of surface refinement and dimensional tolerances. The selected process imposes some constraint in the component's design, such as the inclusion of draft angles, avoiding sharp section changes or sharp radii.

For the functional surfaces of the lower arm the capabilities of the die casting process are not sufficient, so the component must undergo further machining. A boring process can produce appropriately precise feature to fit correctly the bushings and the spherical joint.

The material chosen for the component is Aluminium, in particular the alloy EN AB 46000 which is suitable for die casting: it allows to reach a good compromise between workability, mechanical properties (Yielding strength $YS = 140 \text{ MPa}$) and component's weight.

To properly define the pre-design of the lower arm, its internal actions must be studied to size its cross-sections. The stiffest possible solution would be to connect the points B, C1 and C2 as in the edges of a triangle, but to allow for an outward steering angle of 32° the following design has been adopted:

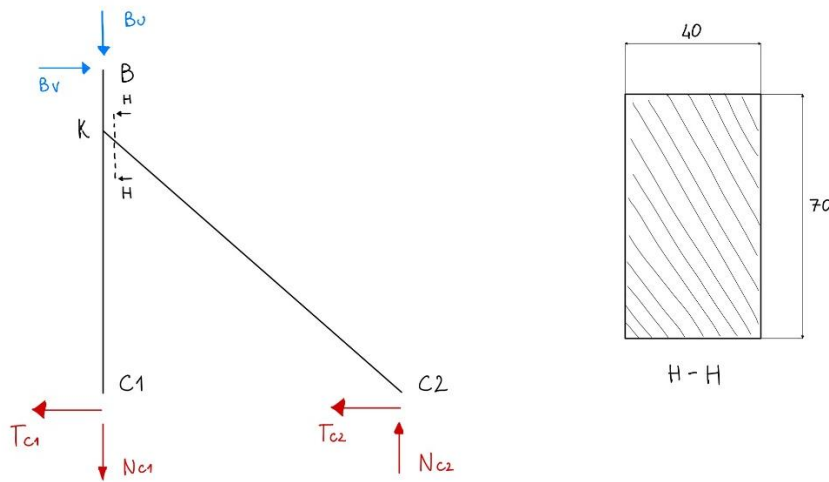


Figure 19: Lower arm's sketch

This exposes the segment BK to a high bending moment, thus making section H-H the most loaded one in the component. With a rectangular section of 70 by 40 mm the resulting equivalent stress in $\sigma_{tot} = 75 \text{ MPa}$, which translates to a safety factor $SF \approx 2$. With this data, a very rough first-try shape was produced:

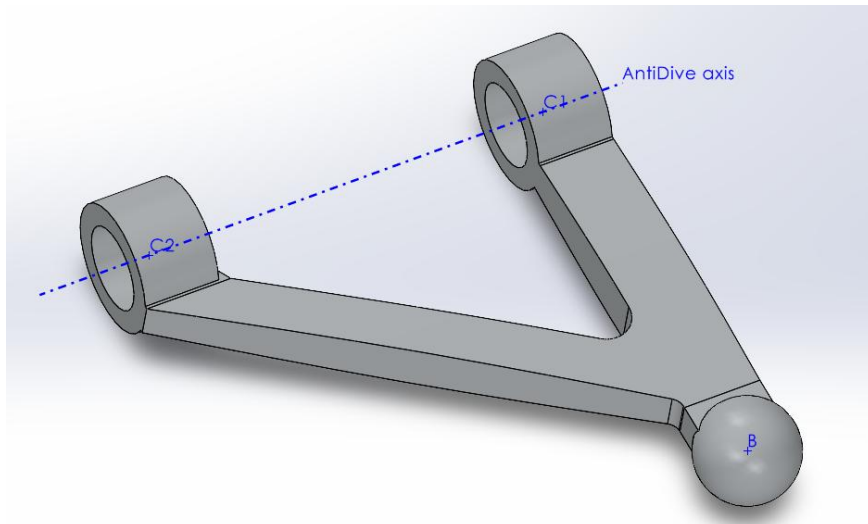


Figure 20: Lower arm's first design

This solution is clearly over-sized, weighting . The component has undergone a topological optimization process thanks to SolidWorks' built-in FEA tool and the output is a much lighter solution, in which the material and its distribution is much better exploited.

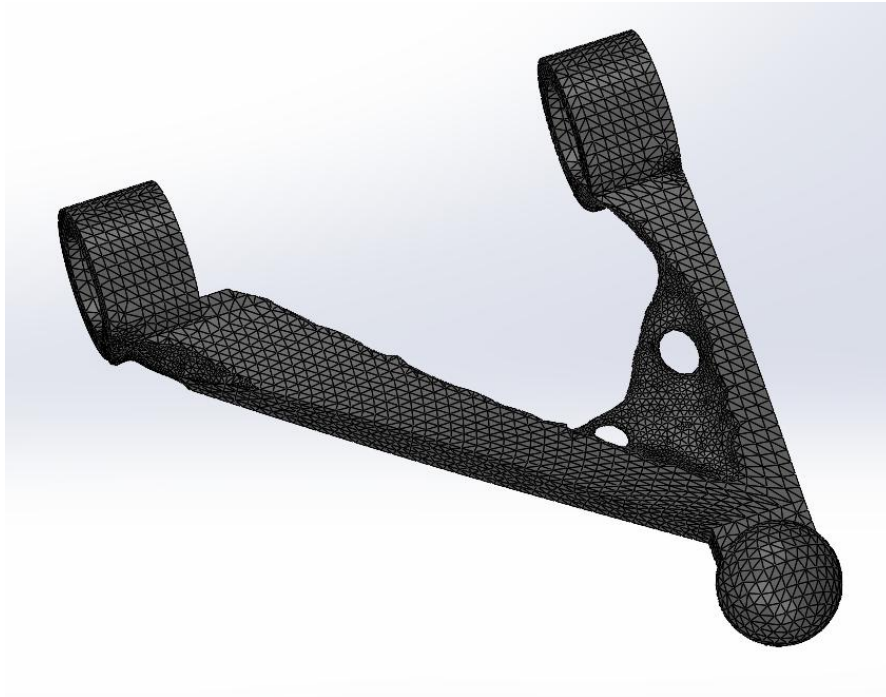


Figure 21: Topological optimization result

This complex shape has no application for a component that must be produced with die casting, but it inspired the lower arm's final design:

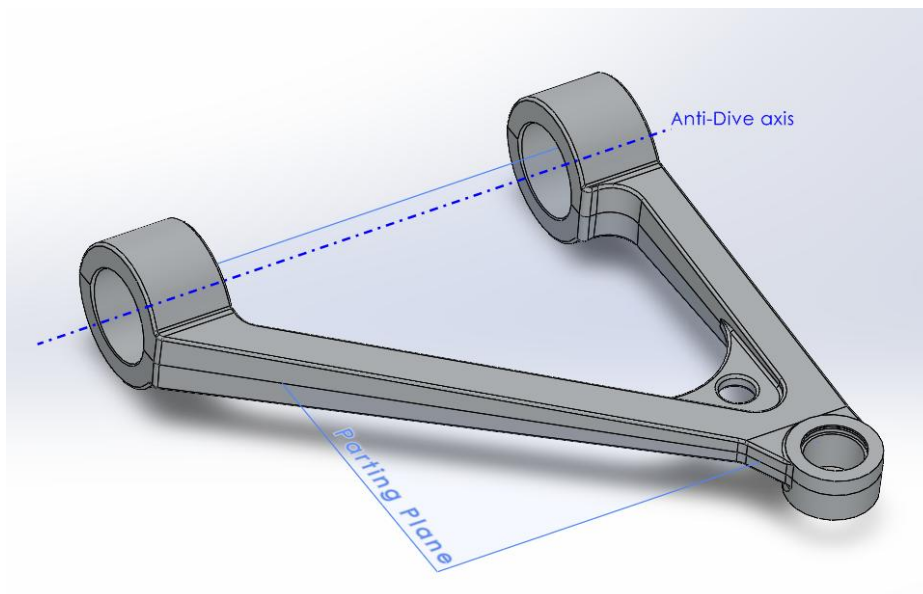


Figure 22: Lower arm's final design

This final iteration shows all the features needed for production (for example draft angles and dies parting plane) and the proper mating surfaces for the bushings and the ball joint assembly. The final weight is 3,51 kg. This design has also been tested with a finite element simulation:

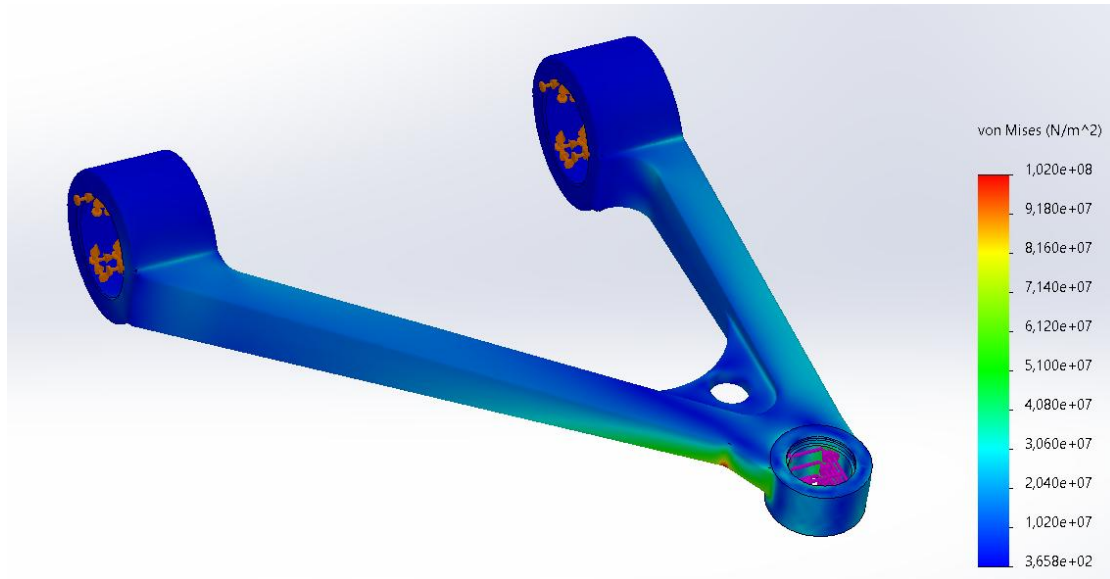


Figure 23: Lower arm finite element analysis

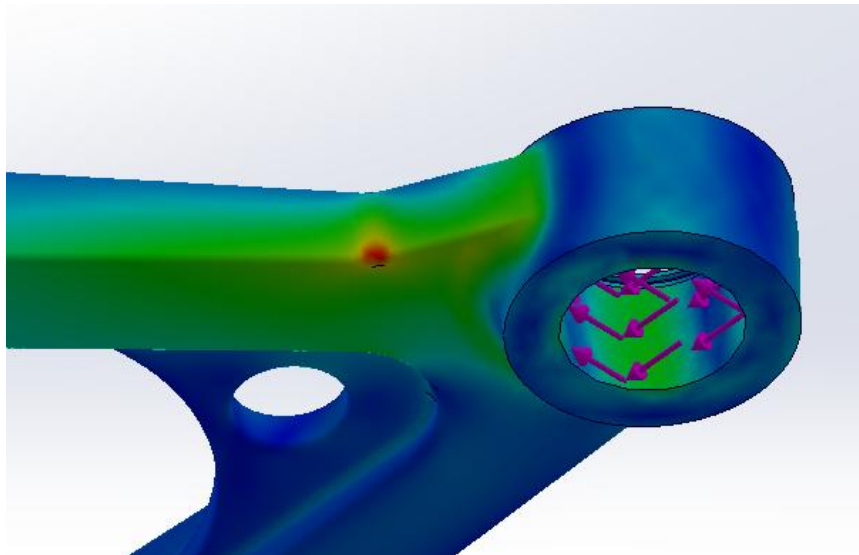


Figure 24: Finite element analysis, most stressed point

As expected, the most critical point is the section previously indicated as H-H (just one side, as the rib makes the other one significantly stronger). The maximum equivalent stress in this point is $\sigma_{tot} = 102 \text{ MPa}$, a value that is coherent with the section reduction resulted by the topological optimization but still well below the yielding strength of the material ($SF = 1,37$). The lower arm can finally be fitted in the suspension assembly:

Suspension assembly

The final stage in the design process involved the creation of a complete suspension assembly, incorporating all essential components and ensuring optimal integration and functionality. The chosen suspension type is the McPherson strut, widely used in modern vehicles due to its simplicity, cost-effectiveness, and efficient use of space.

With the information available and computations performed, the suspension assembly was created using SOLIDWORKS. The designed suspension assembly is illustrated below.

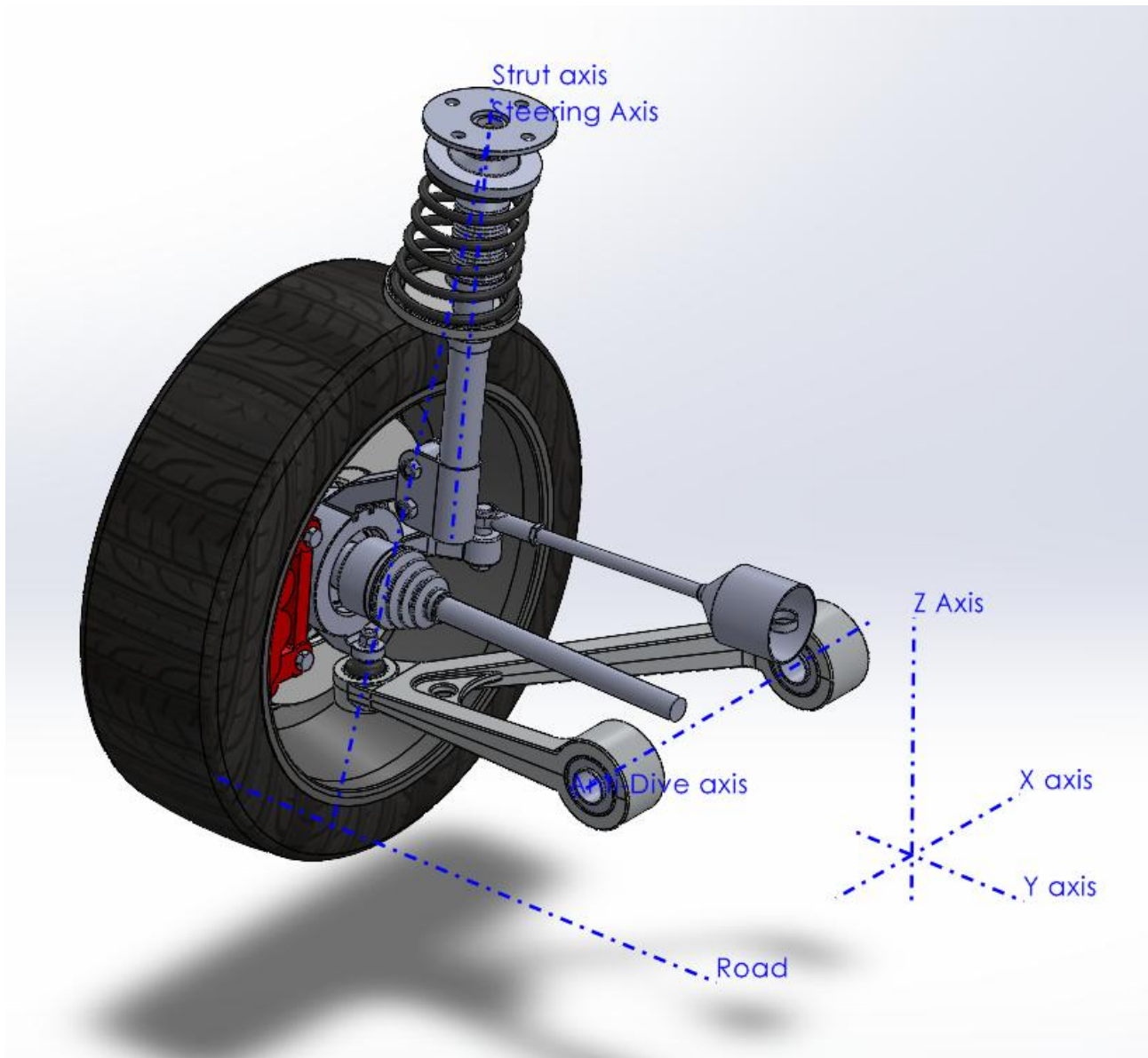


Figure 25: Suspension assembly

Guest lecture on MotoGP braking system by Brembo

Introduction

The motorbike and rider system during the braking phase can be simplified according to the following scheme where are indicated all the dimensions useful to analyse the dynamics of the system.

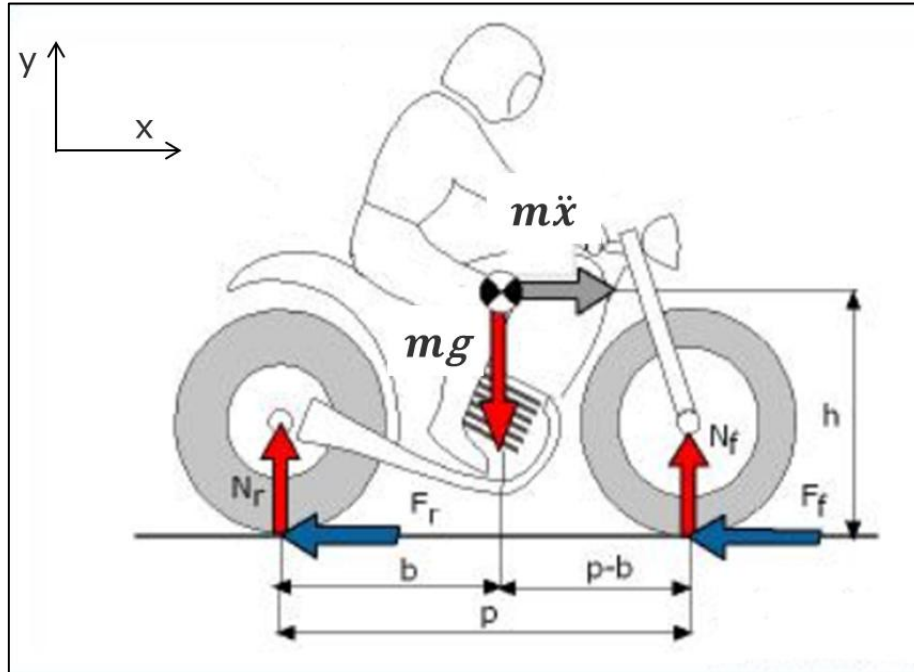


Figure 26: Motorcycle model

Considering the braking phase of motorbike, we must consider the two different phenomena that limit the maximum deceleration. The first is due to the slip limit: since we have a certain friction coefficient between the wheels and the road, the maximum braking force both on front and rear is limited by the maximum friction force that can be generated. The second limit is related to the flipping of the motorcycle, that might happen if the deceleration is too high and consists into the detachment of the rear wheel from the ground, so corresponds to the condition where rear vertical force is null. Depending on motorbike geometry and grip coefficient the deceleration can be limited either by the available grip or by the flipping limit, but it must be said that, for MotoGP's bike, with dry asphalt the flipping limit is the most strict one.

Deceleration analytical description

According to the deceleration also the total braking force can be computed by means of the following formula:

$$F = m * a$$

The horizontal, vertical and moment equilibrium equations can be written for the system and are the following ones:

$$\begin{aligned} \sum F_x &= 0 & m * a - F_r - F_f &= 0 \\ \sum F_y &= 0 & -m * g + N_r + N_f &= 0 \\ \sum M &= 0 & -(F_r + F_f) * h - N_r * b + N_f * (p - b) &= 0 \end{aligned}$$

Now we can write the front and rear vertical forces as function of the braking force.

$$N_f = m * g * \frac{b}{p} + F * \frac{h}{p}$$

$$N_r = m * g * \frac{p - b}{p} - F * \frac{h}{p}$$

If we consider the maximum deceleration to be limited by the grip than it will be:

$$a = g * \mu$$

and the following relation holds:

$$F = F_f + F_r = N_f * \mu + N_r * \mu$$

where μ is the friction coefficient, that in this case is supposed to be the same for front and rear. The maximum deceleration can be achieved with an optimal repartition of the braking force among rear and front, which can be derived by the last equation:

$$\frac{F_f}{F} = \frac{(b + \mu * h)}{p}$$

$$\frac{F_r}{F} = \frac{[(p - b) - \mu * h]}{p}$$

If now we consider the flipping limit the maximum deceleration possible can be derived by imposing the vertical load on the rear to be equal to zero and so also the rear braking force to be null:

$$a = g * \frac{(p - b)}{h}$$

As we can observe the flipping limit deceleration only depends on the geometry.

To conclude if we consider the aerodynamic resistance F_d then the maximum deceleration achievable will be higher:

$$a = g * \frac{p - b}{h} + \frac{F_d}{m}$$

Deceleration results

In this part is presented the solution to the proposed exercise and in the table below are reported all the data considered in this specific case. All the results have been computed according to the formulas previously presented.

m_bike	[kg]	217
m_rider	[kg]	70
m	[kg]	287
p	[mm]	1382
b	[mm]	655
h	[mm]	677
μ	[-]	0.80
ρ	[km/m ³]	1.20
C_d	[-]	0.60
A	[m ²]	0.40
v	[km/h]	200.00

Table 9: Motorbike data

The static load distribution can be easily computed by considering a null braking force acting on the vehicle:

$$\text{Front weight} = 47\%$$

$$\text{Rear weight} = 53\%$$

Then both the deceleration at the flipping limit and at the slipping limit can be computed, considering $\mu = 0.8$ and neglecting the aerodynamic resistance.

$$a_{\text{flipping_limit}} = 1.07 \text{ [g]}$$

$$a_{\text{slipping_limit}} = 0.8 \text{ [g]}$$

As can be noticed the slipping limit is in this case stricter and so the corresponding deceleration is the real maximum one, of course this condition could be changed with a different geometry or a different grip coefficient. Also, the optimal brake repartition to achieve this value of deceleration can be found.

$$\frac{F_f}{F} = 0.87$$

$$\frac{F_r}{F} = 0.13$$

In the next graph we can observe that the optimal braking repartition is related to the vertical load distribution that varies depending on the braking force, so considering the maximum braking force the corresponding load transfer can be determined and so the optimal brake partitioning.

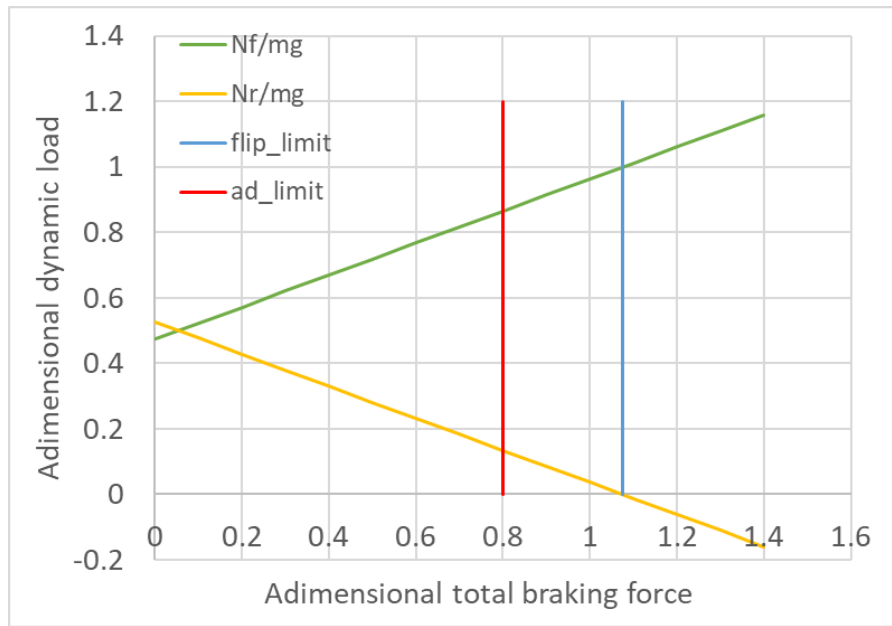


Figure 27: Optimal brake partitioning

What has been said up to now is confirmed by the following graphs, where the braking point is the intersection of the two lines describing the slipping limit, since it is the stricter one. Also, we can observe the curves for different brake partitioning and the iso-deceleration curves. What is clear, again, is that to reach higher deceleration the brake partition must be moved to the front, since the load transfer increases the available braking force on the front while reducing the one at the rear.

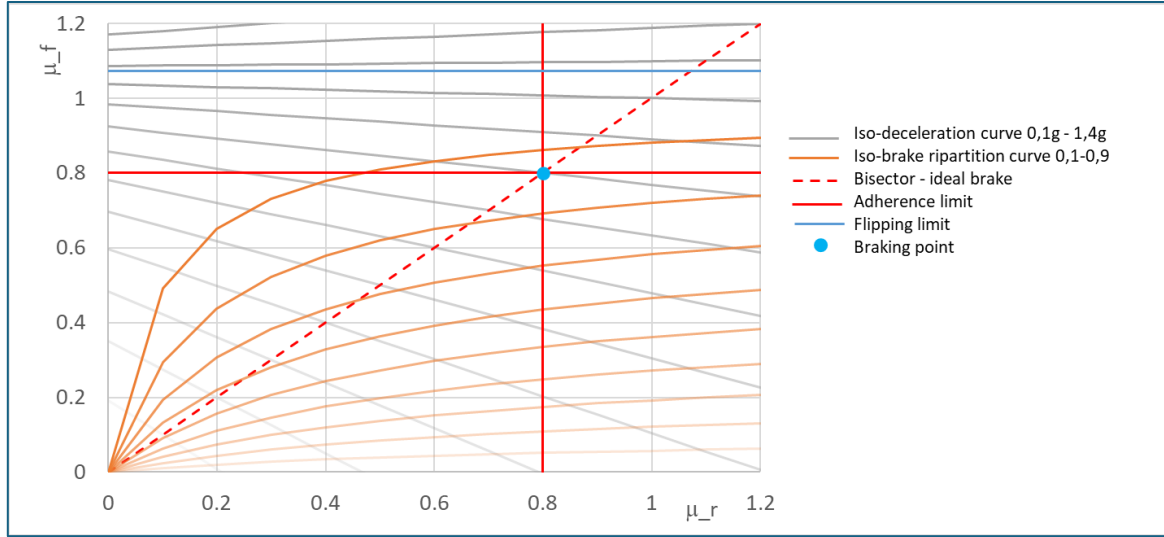


Figure 28: Braking point (friction coefficient graph)

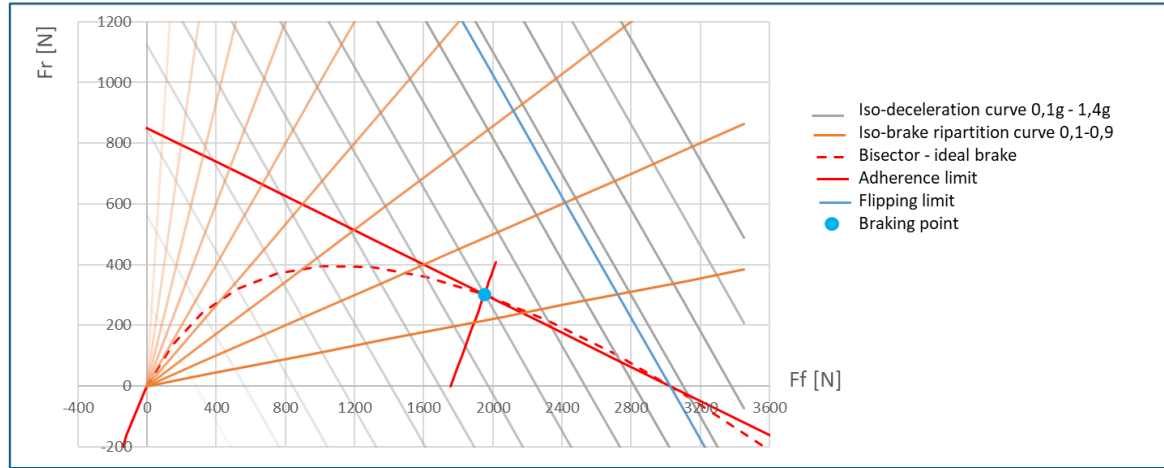


Figure 29: Braking point (braking forces graph)

If we now consider to be braking only with the front or only with the rear the corresponding deceleration values can be found, and, as expected, they are lower than in the case of using both the brakes, but also, due to the load transfer, the deceleration braking with just the rear brake is the lowest one.

$$a_{\text{front braking}} = 0.62 \text{ [g]}$$

$$a_{\text{rear braking}} = 0.30 \text{ [g]}$$

Now, supposing to have enough grip to reach the flipping limit and considering the effect of the aerodynamic resistance the maximum deceleration can be computed.

$$a_{\text{max}} = 1.23 \text{ [g]}$$

Braking system exercise

Considering now to have the same vehicle equipped with 36/38 calipers and a 340 mm diameter disk we can compute the pressure required to achieve the deceleration of the flipping limit. To do this we must know the friction coefficient between the calipers and the disk that in this case is equal to 0.5.

Considering the flipping limit the braking force is equal to the one on the front and the braking torque C can be computed once the tire radius is known.

$$F = F_f$$

$$C = F * r_{\text{tyre}}$$

Now the tangential force on the disk can be computed starting from the braking torque and knowing the disc effective radius, which corresponds to the position where the force is applied on the disk.

$$F_{\text{tan}} = C / r_{\text{disk}}$$

Once the tangential force is known, the normal force can be computed and thus the pressure required.

$$F_{\text{caliper}} = \frac{F_{\text{tan}}}{\mu}$$

$$\text{pressure} = \frac{\frac{F_{\text{caliper}}}{2}}{\pi * \frac{D1^2 + D2^2}{4}} = 2.71 \text{ [bar]}$$

Considering the next scheme for the braking lever and master cylinder the force that must be applied by the driver can be computed.

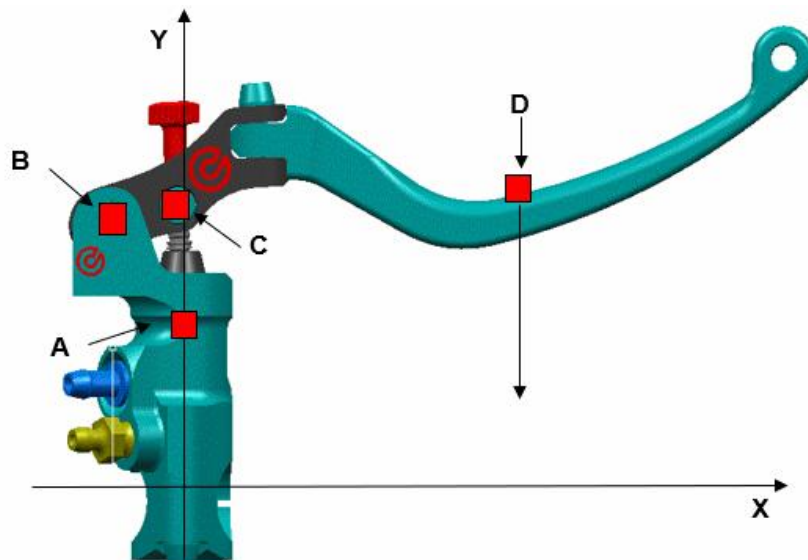


Figure 30: Braking lever - master cylinder scheme

Two different master cylinders have considered to make a comparison: 19x18 and 19x20; the first number is cylinder diameter while the second one is the position of point B with respect to the cylinder axis. Then a distance equal to 150 mm have been considered between point B and point D, which is where the force is applied by the driver. The force can be easily computed according to the following formula:

$$F_{\text{rider}} = \text{pressure} * \pi * \frac{d^2}{4} * \left| \frac{x_b}{x_d - x_b} \right|$$

So, in the two different cases the forces will be:

$$F_{\text{rider19x18}} = 92.2 \text{ [N]}$$

$$F_{\text{rider19x20}} = 102.4 \text{ [N]}$$

As we see the force to be applied depends by the mechanical ratio of the master cylinder, thus this parameter is very important, since it is strictly related with the feeling of the driver and so it is fundamental to achieve the best performance.

Visit to Pirelli test track in Vizzola

Introduction:

The visit to Circuito Vizzola and the guest lecture from Pirelli provided a unique and valuable opportunity to delve into the complexities of automotive tire performance and testing. As a global leader in the tire industry, Pirelli's demonstration and explanations offered in-depth knowledge of how tires are evaluated under various conditions to ensure safety and performance. This report details the comprehensive teachings and learnings from the visit, focusing on the three main testing components: Aquaplaning, Handling, and Steering. It further elaborates on the practical applications of these insights and how they contribute to our understanding of tire dynamics and vehicle safety.

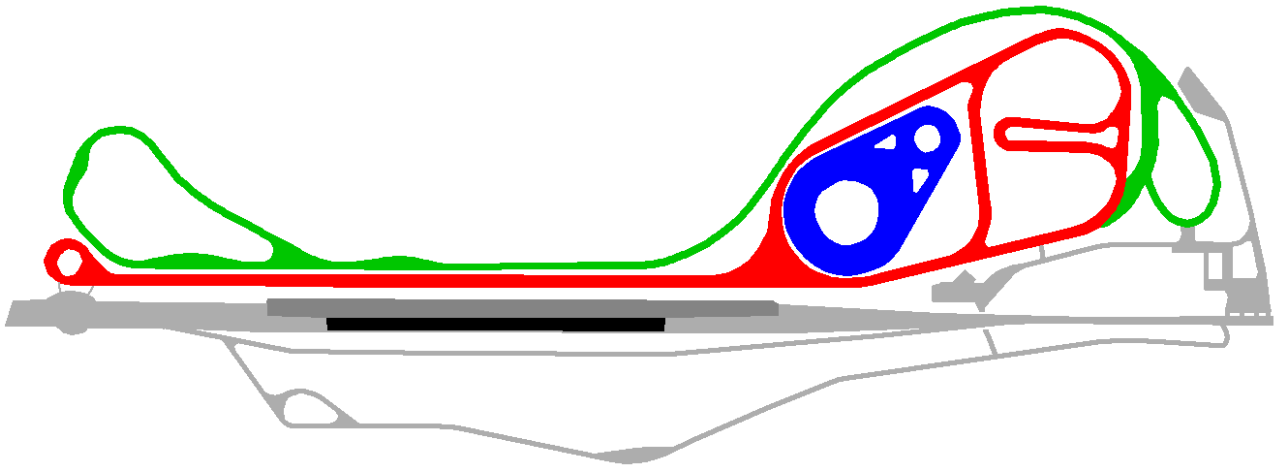


Figure 31: Pirelli test track map

The session began with an overview of Pirelli's history, customer base, and various activities. This introductory segment highlighted Pirelli's commitment to innovation, quality, and performance, setting the context for the detailed technical demonstrations that followed. The practical demonstrations were structured around three primary testing components:

1. Aquaplaning
2. Handling
3. Steering

Aquaplaning test

In the aquaplaning test, an Audi A4 was driven on a straight track intentionally flooded with water to a depth of 7mm. The goal was to observe the car's steering behaviour at different speeds:

1. First Round (60 kmph): At 60 kmph in 4th gear, the car exhibited minimal steering response. This slight response indicated that while some control is retained, it is significantly reduced in water-filled conditions.
2. Second Round (70 kmph): At 70 kmph in 4th gear, the car failed to steer entirely when attempting to manoeuvre through the water. This means that the front tyres have completely lost contact with the ground. This stark contrast in behaviour between the two speeds illustrated the critical threshold at which aquaplaning becomes a severe risk, emphasizing the necessity of lower speeds (around 30 kmph) for effective steering and control in such conditions.

Handling test

The handling test was conducted using a BMW 3 Series on a wet track designed with various road structures, including turns, climbs, and flat sections. Cones were arranged to simulate different driving scenarios:

The test showcased how the vehicle manages these diverse conditions, highlighting the importance of tire performance in maintaining stability and control. The experience of driving through different sections

demonstrated the dynamic response of the tires to wet conditions, stressing the need for precise tire engineering to ensure consistent performance and safety.

Steering pad test

The steering test also utilized a BMW 3 Series and took place in the steering pad zone of the track:

1. Nominal Speed Test: At nominal speed and with gradual steering input, the car maintained its course without oversteering, showcasing the typical behaviour expected from a rear-wheel-drive (RWD) vehicle under controlled conditions.
2. Sudden Input Test: When sudden acceleration and steering input were applied, the car oversteered, contrary to the expected behaviour indicated by the handling diagram. This oversteer occurred because the rear tires lost grip more quickly than the front tires, (the rear effective axle characteristic curve to rose above the front one in the handling diagram). This behaviour, typical of RWD cars under high cornering forces or rapid acceleration, provided a practical demonstration of the dynamic limits and characteristics of RWD vehicles. The driver intervention in this situation was fundamental to stabilize the car, applying rapid and precise countersteer to counteract the yaw motion of the car.

Learning and outcomes

The aquaplaning test highlighted the significant impact of speed on vehicle control in water-filled conditions. The inability to steer at higher speeds underscored the importance of maintaining lower speeds to prevent loss of control due to aquaplaning. This understanding is crucial for both vehicle design and driver safety training.

The handling test clearly showed how wet conditions and various road structures affect tire performance and vehicle stability. Observing the car's response to turns, climbs, and flat sections in wet conditions emphasized the need for rigorous tire testing and the development of tires that can maintain optimal performance across diverse driving scenarios. Moreover, it offered a valuable opportunity to observe vehicle dynamics firsthand under low-grip conditions. Maneuvers such as hard braking and rapid acceleration demonstrated the critical role of precise driving inputs in maintaining control. It became evident that a solid understanding of driving techniques is essential for maximizing vehicle performance and ensuring safety.

The steering test offered valuable insights into the steering dynamics of RWD vehicles. The demonstration of oversteer when sudden inputs were applied illustrated the complex interactions between acceleration, steering, and tire grip. Understanding these dynamics is essential for designing safer and more responsive vehicles and developing advanced driver assistance systems (ADAS) that can mitigate the risks associated with sudden manoeuvres.

Conclusions

The guest lecture and visit to Circuito Vizzola-Pirelli provided an helpful learning experience, offering practical insights into tire testing and vehicle dynamics. The aquaplaning, handling, and steering tests demonstrated the critical importance of tire performance in various conditions and the need for meticulous testing to ensure safety and efficiency. The teachings from this visit emphasized the significance of speed control in preventing aquaplaning, the dynamics and handling in wet conditions, and the steering behaviour of RWD vehicles under different inputs. These learnings are instrumental in advancing our understanding of tire and vehicle dynamics, contributing to developing safer and more efficient automotive technologies.

Design of light-weight aluminium wheels

Introduction

The focus of this exercise is the design and assessment of lightweight aluminium wheels for the BMW 330i 2020 model, emphasizing safety, performance, and style. This exercise involves computing the fatigue life of an aluminium alloy wheel subjected to a dynamic cornering fatigue test, based on ISO and SAE standards.

The process begins with computing the bending moment for the test, followed by calculating the Sines fatigue stress and determining the fatigue life of the wheel. Key methodologies include the use of digital twins for accurate modelling and simulation of physical phenomena, enabling lightweight and safe designs while minimizing costs and unexpected failures. This assessment encompasses various safety tests such as ISO 3006, SAE J328, and SAE J2562 for fatigue, as well as SAE J175 and ISO 7141 for impact. The dynamic cornering fatigue test involves determining the bending moment using parameters such as friction coefficient, tire radius, and vertical static load. Additionally, the exercise employs finite element modelling to simulate dynamic stress states and combines them to represent the rotating bending moment. The Sines fatigue criterion and Wohler's curve are used for fatigue life estimation, utilizing MATLAB for stress tensor component analysis.

By conducting this comprehensive analysis, this test aims to ensure the durability and reliability of aluminium wheels under various loading conditions, providing valuable insights into their performance and contributing to the overall safety and efficiency of the BMW 330i.

The standard tire used in BMW 330i as specified by the manufacturer is 225/50 R 17 98Y. The load index and speed rating for these tires are critical for ensuring both safety and performance under various driving conditions. The load index for the specified tire is 95, which indicates that each tire can support a maximum load of 750 kg. This value is crucial for determining the overall load-carrying capacity of the vehicle, ensuring that the tires can handle the weight of the vehicle, passengers, and any additional cargo without compromising safety. The speed rating for this tire is Y, which corresponds to a maximum safe speed of 300 km/h. This rating is essential for understanding the safe operational limits of the tires, ensuring they can handle the stresses and heat generated at higher speeds without failure. The load rating and speed index values have been referenced from the wheel-size.com website, which provides comprehensive and reliable information on tire specifications.

Wheel durability assessment

The bending moment is crucial for simulating the loads experienced by a wheel during cornering and establishing the appropriate test conditions to evaluate its durability. The bending moment (M) acting on the wheel during the test is calculated using the formula:

$$M = (\mu R + d) \cdot F_y \cdot S$$

Where:

- M : Bending moment (Nm)
- μ : Assumed friction coefficient between tire and road (0.7)
- R : Tyre radius (m)
- d : Inset or outset of the wheel (m)
- F_y : Maximum vertical static load of the vehicle or the largest value of the load rating of the tires mounted on the wheel (N)
- S : Accelerated test factor

The dynamic cornering fatigue test is conducted using a finite element model to ensure the wheel's durability under cornering loads. The model employs a linear elastic material with an elastic modulus (E) of 70 GPa and a Poisson's ratio (ν) of 0.3. The force application is facilitated through a rigid link to accurately simulate real-world conditions. The finite element model utilizes quadratic tetrahedral elements with a typical dimension of 8 mm, refined locally down to 4 mm to capture the detailed stress distribution and deformation accurately. This approach allows for a comprehensive evaluation of the wheel's performance under dynamic cornering forces, ensuring reliability and safety in practical applications

To approximate the complex stress state under a rotating moment, the stress states resulting from two orthogonal unitary bending moments are superposed. The formula used for dynamic stress computation is:

$$\sigma(\alpha) = M(\sigma_{M_z} \cos \alpha + \sigma_{M_x} \sin \alpha)$$

where (α) represents the rotation angle. This method allows for an accurate assessment of the component's stress response and fatigue life under dynamic loading conditions. The provided methodology outlines a fundamental approach to fatigue life estimation using the Sines fatigue criterion and the Wohler curve. The Sines fatigue criterion calculates the equivalent stress amplitude (σ) from the principal stress amplitudes and shear stress amplitudes, as expressed by the formula:

$$\sigma_S = \sqrt{\sigma_{x,a}^2 + \sigma_{y,a}^2 + \sigma_{z,a}^2 - (\sigma_{x,a}\sigma_{y,a} + \sigma_{x,a}\sigma_{z,a} + \sigma_{y,a}\sigma_{z,a}) + 3(\tau_{xy,a}^2 + \tau_{xz,a}^2 + \tau_{yz,a}^2)}$$

This equivalent stress is then used with the Wohler curve to predict the fatigue life (N_f). The Wohler curve, a logarithmic representation of stress amplitude versus the number of cycles to failure, determines the fatigue life corresponding to the calculated equivalent stress using the equation:

$$\log(\sigma_{lim}) = \log(\sigma'_f) + b \log(2N_f)$$

The slope of the Wohler curve, represented by the coefficient 'b', characterizes the material's fatigue behaviour. This method provides a foundational approach to fatigue life estimation; however, actual fatigue life prediction often involves more complex considerations, such as mean stress effects, load spectrum, and material behaviour under multiaxial loading conditions.

The methodology for fatigue life assessment involves two main steps: first, calculating the stress state at nodal points by superimposing stress contributions from two orthogonal unitary bending moment load cases; second, predicting fatigue life by calculating the equivalent stress amplitude using the Sines criterion and employing the Wohler curve parameters such as slope (b) and fatigue strength coefficient. These computations are conducted using MATLAB, ensuring a robust analysis of the component's durability under varied loading scenarios. This approach accounts for material fatigue characteristics and stress distribution within the component, providing a comprehensive assessment of its fatigue life expectancy.

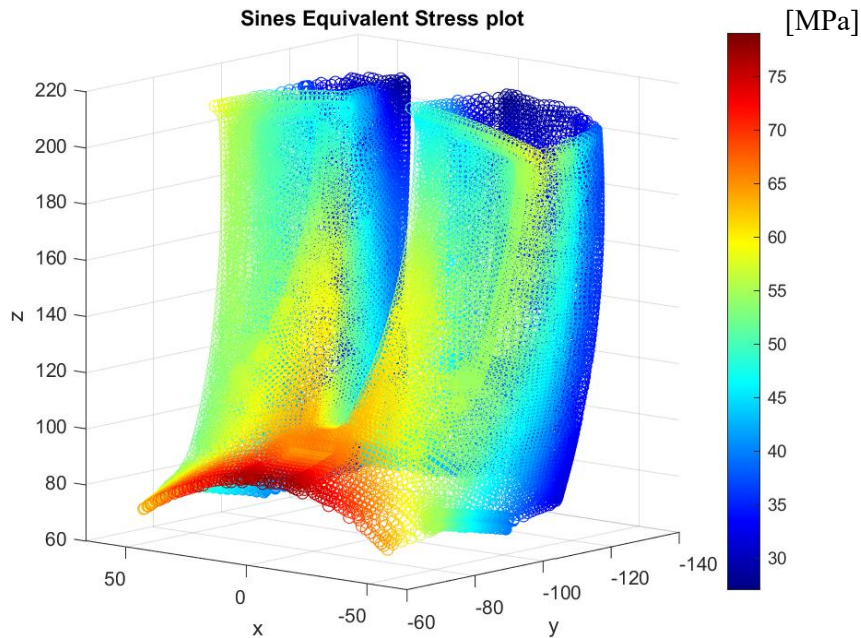


Figure 32: Sines equivalent stress plot

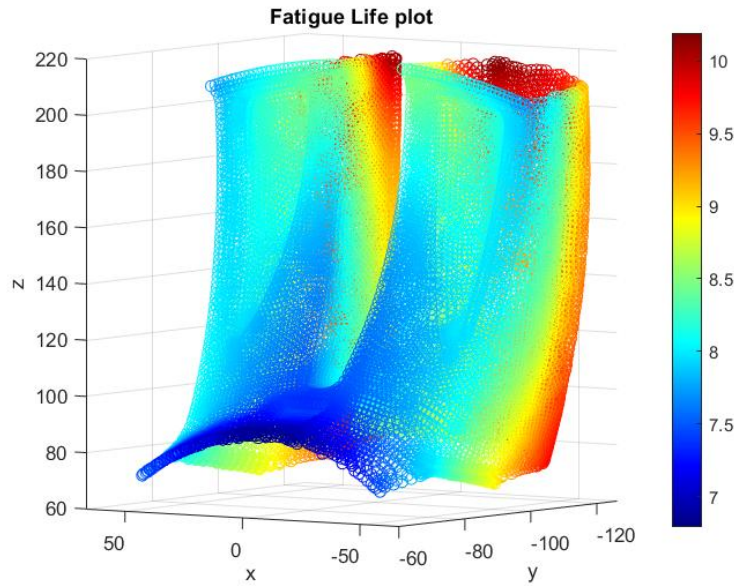


Figure 33: Fatigue life plot

The fatigue assessment of the wheel illustrated through Sines equivalent stress and fatigue life maps, identifies critical stress concentrations and potential failure points. High Sines equivalent stress regions, marked in red, are concentrated at the close to the inner mating surface of the wheel, corresponding to areas with the shortest fatigue life. These high-stress zones are most susceptible to fatigue failure under cyclic loading. Conversely, the outer portion of the wheel, exhibiting lower stress levels and longer fatigue life, is less prone to failure. These insights highlight the necessity for targeted design enhancements or reinforcements at the outer edges and inner corners to improve the wheel's overall durability and service life.

These high-stress zones are most susceptible to fatigue failure under cyclic loading, aligning with the ISO 3006 standard for road vehicle wheels and the dynamic cornering fatigue test methodology, which evaluates the wheel's performance under simulated bending moments and stress states. Where the minimum number of fatigue cycles the wheel has to sustain to pass the test is 50,000 for an Accelerated test factor (S) of 2 for Aluminium and our rim at the weakest point sustains for about 10^7 cycles. The stress computation involved simulating two orthogonal unitary bending moments combined and scaled to represent the rotating bending moment. The Sines fatigue criterion was applied to estimate fatigue life, incorporating stress tensor components and Wohler's curve parameters for the material. The central portion of the wheel, displaying lower stress levels and longer fatigue life, is less prone to failure, indicating effective load distribution and structural integrity in this area. These findings underscore the necessity for targeted design enhancements or reinforcements at the outer edges and inner corners to improve the wheel's overall durability and service life. By addressing these critical stress zones, it is possible to enhance the wheel's performance, ensuring it meets safety standards and performs reliably under real-world conditions.

Guest lecture on aluminium alloy wheels by Cromodora wheels

Introduction

The guest lecture was focused on "Aluminium Alloy Wheels at Cromodora Wheels." The session was conducted by C. Sorlini, providing a comprehensive overview of Cromodora Wheels' operations, aluminium alloy wheel's types, manufacturing processes, and the company's quality planning and research initiatives. This report encapsulates the teachings and learnings from the lecture, emphasizing the key points discussed.

Company overview

Cromodora Wheels has a rich history dating back to its foundation in 1928. Significant milestones include producing the first high-pressure magnesium wheel in 1966 and the start of aluminium alloy wheel production in 1969. Over the decades, the company has undergone several ownership changes and expansions. Currently, Cromodora operates two plants—one in Ghedi, Italy, and the other in Mosnov, Czech Republic.

The Ghedi plant covers 60,000 sqm with a focus on wheels up to 23 inches, exotic finishes, and advanced machining capabilities. The Mosnov plant, larger at 96,000 sqm, specializes in high production runners and automation, with a nominal capacity of 2 million units. Both plants boast state-of-the-art equipment for foundry, machining, painting, and final inspection, ensuring high-quality production standards.

Cromodora Wheels actively engages in research and development, proposing internships and projects to foster innovation. The company's commitment to R&D is evident in its continuous process improvements and the introduction of new technologies such as flow forming and advanced machining techniques.

Aluminium alloy wheel types

- **Cast Wheels:** they are the most common type, produced through low-pressure die casting (LPDC) or gravity die casting (GDC). LPDC involves molten metal filling the die from the bottom, while GDC relies on gravity to fill the die. Both processes allow for a wide variety of designs at reasonable costs.
- **Forged Wheels:** they offer superior material characteristics and lower weight compared to cast wheels. They can be net forged or blank forged and subsequently machined. Net forging is cost-effective but limits design variety, whereas blank forging allows for intricate designs suitable for high-performance vehicles.

Wheel main features

- **Geometrical Characteristics:** Key geometrical features of wheels include style, diameter, rim width, pattern circle diameter (PCD), offset, and hump and rim type. These features are crucial for ensuring proper fitment and performance.
- **Interfaces:** Wheels must interface correctly with various vehicle components. Important interface characteristics include tire types and sizes, brake calliper packaging profiles, hub diameter, centre cap design, and valve placement.
- **Cosmetics and Dynamics:** Cosmetic aspects such as colours and finishes (e.g., fully painted, diamond-turned) play a significant role in the aesthetic appeal of wheels. Dynamic features include vertical and lateral load capabilities, natural frequencies, and stiffness, which are essential for safety and performance.
- **Functional Specifications:** Functional specifications encompass mechanical, metallurgical, and corrosion resistance tests. These tests ensure the wheels meet stringent quality standards for durability and performance. Functional tolerances also include limits on dynamic unbalance to maintain smooth operation.

Advanced Product Quality Planning (APQP)

APQP is a structured framework used predominantly in the automotive industry to ensure product quality from conception to production. It comprises five phases:

1. Bid/Win: feasibility studies, pitching to customer, defining the business case.
2. Product Design: Design and dimensioning, ending with a design freeze.
3. Product Tooling and Validation: Tooling, sampling, and validation, leading to customer engineering approval.
4. Product Approval for Production: Pre-series production, quality studies, and customer quality approval.
5. Production Ramp-Up: Full production capacity installation and performance measures, ending with project handover to the production plant.

Learning and outcomes

The guest lecture provided an in-depth understanding of the operations and technical aspects of aluminium alloy wheel production at Cromodora Wheels. Several key learnings and outcomes emerged from this lecture:

Manufacturing Processes:

1. Low-Pressure Die Casting (LPDC): Understanding the mechanics of LPDC helped us appreciate the benefits of producing complex designs at lower costs.
2. Forging Techniques: Insights into net forging and blank forging highlighted the trade-offs between cost, weight, and design flexibility.

Quality Assurance:

1. The rigorous testing methods employed by Cromodora, including mechanical, metallurgical, and corrosion resistance tests, underscored the importance of maintaining high standards for durability and performance.
2. APQP's structured approach to quality planning provided a clear roadmap from project initiation to full-scale production, ensuring each phase meets predefined quality benchmarks.

Design and Interface Considerations:

1. Geometrical and interface characteristics such as rim width, PCD, and brake calliper packaging are critical for ensuring compatibility and performance.
2. Cosmetic features and dynamic properties significantly influence customer satisfaction and vehicle handling characteristics.

R&D and Innovation:

1. Cromodora's focus on continuous improvement and the adoption of cutting-edge technologies like flow forming demonstrates the company's commitment to innovation and staying ahead in a competitive market.

Conclusion

The guest lecture offered a valuable perspective on the complexities of wheel manufacturing and the stringent quality control measures necessary for producing high-performance aluminium alloy wheels. It highlighted Cromodora Wheels' robust processes and commitment to innovation, quality, and customer satisfaction. These insights not only broadened my understanding of the technical and operational aspects of wheel production but also reinforced the importance of continuous learning and adaptation in engineering practices. This knowledge is crucial for evaluating and improving vehicle components, ultimately contributing to the advancement of automotive engineering.

Guest lecture on functional safety and design FMEA by Capgemini

Introduction

Capgemini conducted a comprehensive guest lecture focusing on two critical aspects of engineering: Functional Safety lectured by Elena Caovilla and Design Failure Modes and Effects Analysis (FMEA) lectured by Elisa Compierchio. The lecture was designed to provide in-depth knowledge and practical insights into these essential topics, widely applicable in various industries, particularly automotive and aerospace. The following sections provide a detailed summary of the teachings and learnings from the lecture, divided into two main parts: Functional Safety and FMEA, each followed by reflections on the learnings and outcomes.

Functional Safety

Functional Safety is a discipline that ensures systems operate correctly in response to their inputs, even when there are potential failures. It is particularly crucial in applications where malfunctions can lead to severe consequences, such as in the automotive, medical, and industrial sectors.

The lecture began with an introduction to functional safety, emphasizing its importance in ensuring that systems perform reliably and safely, especially in critical applications. The concept of functional safety revolves around preventing hazards caused by system failures. This involves designing systems to handle faults predictably, ensuring minimal risk to users and the environment. **Hazard analysis and Risk Assessment (HARA)** are foundational activities in functional safety. These processes identify potential hazards, assess their risks, and define measures to mitigate these risks to an acceptable level. Key metrics include the severity, exposure, and controllability of potential hazards. The HARA process ensures that all potential risks are systematically identified and managed, reducing the likelihood of unexpected failures.

Safety functions are specific actions taken by the system to avoid hazardous situations. The lecture detailed how to define **safety functions** and allocate a **Safety Integrity Level (SIL)** to each, which quantifies the required reliability of these functions. SIL ranges from 1 to 4, with SIL 4 representing the highest level of safety integrity. This classification helps in designing systems that meet the required safety performance, ensuring that critical functions are reliable and effective. The complete **lifecycle of functional safety**, from concept through decommissioning, was outlined. Each phase of the lifecycle was discussed in detail, highlighting the need for continuous monitoring and verification to ensure that safety requirements are met throughout the system's operational life. This lifecycle approach ensures that safety is integrated into every product development and maintenance stage. The international standard **ISO 26262**, which provides a framework for functional safety in the automotive industry, was thoroughly examined. This standard specifies requirements for all aspects of functional safety, including development, production, operation, service, and decommissioning. The standard helps ensure that automotive systems meet stringent safety requirements, enhancing the overall reliability and safety of vehicles.

Failure Modes and Effect Analysis (FMEA):

FMEA is a systematic method for identifying and addressing potential failure modes within a system. It is used to enhance the reliability and safety of products by analysing possible points of failure and their effects on the overall system. The lecture introduced the FMEA methodology, explaining its purpose and importance in product development. FMEA helps in identifying potential failure modes,

understanding their causes and effects, and prioritizing actions to mitigate them. This method is crucial for improving product design and ensuring that potential issues are addressed early in the development process.

Types of FMEA:

1. **Design FMEA (D-FMEA):** Focuses on potential failures related to product design. It aims to improve the design to prevent failures and enhance reliability. D-FMEA analyses various design aspects to identify weaknesses and recommend improvements, ensuring the product meets its intended functions under all conditions.

2. Process FMEA (P-FMEA): Examines potential failures in manufacturing and assembly processes. It aims to identify and mitigate process-related risks. P-FMEA helps in optimizing manufacturing processes to minimize defects and improve quality, ensuring that the final product is reliable and safe.

The step-by-step process of conducting an FMEA was outlined:

1. Identify potential failure modes for each component.
2. Determine the effects of each failure mode on the system.
3. Analyse the causes of each failure mode.
4. Assess the severity, occurrence, and detectability of each failure mode to calculate the **Risk Priority Number (RPN)**.
5. Prioritize failure modes based on RPN and implement corrective actions to mitigate high-priority risks.

The lecture included a detailed case study of a front bumper design, illustrating the application of D-FMEA. The analysis covered potential failure modes related to mechanical stress, temperature variations, and crash absorption. A functional correlation matrix and risk matrix were used to identify and prioritize risks, followed by recommendations for design improvements.

Learning and outcomes

The functional safety section of the lecture provided a robust framework for understanding and implementing safety measures in engineering systems. Key takeaways include:

1. Hazard analysis and risk assessment are critical in identifying and mitigating potential system failures.
2. The importance of defining safety functions and assigning appropriate SIL levels to ensure system reliability.
3. Understanding the lifecycle approach to functional safety emphasizes the need for ongoing evaluation and improvement.

These insights are significant for designing systems that are not only functional but also safe and reliable, reducing the risk of failures that could have severe consequences.

The FMEA segment of the lecture highlighted the importance of a structured approach to failure analysis. Key learnings include:

1. Understanding the methodology and steps involved in conducting an FMEA.
2. The ability to identify and prioritize potential failure modes and their effects on the system.
3. The practical application of FMEA in real-world scenarios demonstrates how systematic analysis can improve product safety and reliability.
4. These learnings underscore the importance of FMEA in proactive risk management and continuous improvement of product design and manufacturing processes.

Conclusion

The guest lecture by Capgemini on Functional Safety and FMEA provided a comprehensive understanding of these critical topics. Through detailed explanations, practical examples, and case studies, the lecture highlighted the importance of systematic analysis and proactive measures in ensuring system safety and reliability. The knowledge gained from this lecture is invaluable for anyone involved in engineering design and development, equipping them with the tools and methodologies needed to create safer, more reliable products. The emphasis on continuous improvement and risk management will undoubtedly influence future work, leading to enhanced product performance and safety.

LaST visit

The LaST (Laboratorio Sicurezza Trasporti) at Politecnico di Milano is dedicated to advancing the safety of various transportation systems. Researchers at LaST focus on minimizing injury risks for occupants of vehicles such as cars, trains, and helicopters. Their work encompasses both passive and active safety measures, aiming to improve overall transport safety through innovative research and development. We were given the opportunity to visit LaST and see in person various examples of the work that is being carried out.

Various types of suspensions were exposed. An Alfa Romeo Giulia was lifted to show its refined systems: a double-wishbone with virtual joint of the lower arm at the front and a very sophisticated multi-link at the rear, with multiple-stage vibration filtering. It was possible to see a more conventional layout on an Opel crossover, with Mac-Pherson at the front and torsion beam at the rear. Finally, the rear axle assembly of an Acura was on display: it was possible to see clearly all the features of its multi-link suspension and the peculiar behaviour of its electronically controlled differential. In addition to this, several components were on view: suspension arms, synchronizers, differentials, hubs, hub carriers among others. This allowed to study in detail their function in the vehicle and their possible manufacturing processes.

Among LaST equipment it was also possible to see the InTenso test rig: a carrying system (steel cage), which bears the body under test and measures the inertia properties of the studied vehicle (inertia tensor and centre of gravity position) using load cells and encoders. Another remarkable instrument is Ruotavia: a horizontal steel drum apparatus useful to design and test road tyres. Lastly, the lab offers another way to measure and assess tyre-road contact forces: a purpose-designed wheel rim equipped with many extensometers can measure the forces going through the wheel.

The visit to the LaST laboratory was valuable, providing a hands-on experience with various components and an opportunity to observe their functionality. It was particularly interesting to see the advanced tools and equipment used for design and development in the automotive sector. This firsthand exposure to real components and innovative research methods significantly enhanced our understanding of the complexities involved in ensuring transportation safety.

DriSMi visit

The DriSMi (Driving Simulator Politecnico di Milano) is an advanced driving simulator located at the Bovisa campus of the Politecnico di Milano. Adopted in February 2021, it is a DiM400 DYNAMIC driving simulator powered by VI-grade simulation software and it is utilized for a wide range of studies on vehicles and their subsystems. Its applications include the design of new materials and components, ride and handling assessments, real driving emission applications, active assistance systems (ADAS), and autonomous and connected vehicles. The simulator features innovative technologies and redundant degrees of freedom for accurate separation between low and high-frequency motions, enabling Software and Hardware-in-the-Loop (SiL and HiL) validations. Equipped with a realistic driving cabin, immersive audio and video systems, and a variety of stimuli, the DriSMi provides a highly realistic driving experience, making it a key resource for driver training and road safety improvement, particularly in the fields of trucks, motorcycles, and high-speed cars. The offered experience consisted in running a brief path on a straight road with consecutive bumps, in this way it has been possible to feel directly the behaviour of the car respect to high and low frequency vibrations.

As part of the lab visit, we had the privilege of testing the POLIMI driving simulator. The objective of this experience was to feel the effect on comfort of the main two vertical vibration modes of a car. First, a ride over road irregularities at a pure frequency of 1 Hz (lower frequency mode). This scenario felt relatively sustainable, with minimal discomfort observed. It was fascinating to note how our eyes instinctively stabilized the view, effectively mitigating the impact of the vibrations. Following this, we experienced a simulation at pure 10 Hz frequency, where the vibrations were more evident and made the discomfort much more apparent, offering a clear understanding of how frequency plays a critical role in the perception of ride quality. The visit concluded with a simulation of driving over virtual speed bumps, which caused in the car a vibration that should be the superposition of the two vibration modes.

Overall, this experience provided a firsthand perspective on the interaction between the vehicle's suspension system and uneven surfaces, reinforcing the importance of suspension design in optimizing comfort and safety. It was quite difficult to distinguish between the two modes when riding over the speed bumps, mainly because of the short duration of the simulation. In that regard, the explanation given by the DriSMi staff and their answer to our question were essential to better understand the experience.